Going from 1+1=2 to Quantum Mechanics

Chaidhat Chaimongkol

May 19, 2026

Hi, my name is Chai and I'm taking an undergraduate quantum mechanics course at UCLA.

It's known as one of the hardest classes at UCLA.

This text is written so that, at the start, you only need to know how to do basic middle school arithmetic.

In the end, you'll know how to do quantum mechanics problems.

This text is loosely sorted by complexity. Everything is derived from concepts introduced in previous sections. As we introduce mathematical concepts, we will use them to better understand quantum mechanics.

This is based on the Electrical and Computer Engineering 128 course taught at the University of California, Los Angeles by Dr. Leo Zhou.

None of this is written by AI

None of this is written by AI — I am writing this by hand because I find explaining concepts makes me understand them better too.

Everything must be defined above

Any mathematical term MUST be derived from first principles (or like elementary school math) in a section before it.

Example

invariant subspace → subspace → subset → set theory

chaimongkol@ucla.edu

Please let me know at chaimongkol@ucla.edu on how I'm doing or that you've even read this message. If I see support, it makes me want to continue this series! The series is currently incomplete.

Contents

1 Let

Ok when I say "let," i'm just defining something to be another thing.

Example:

Let blorple be blue

Now if i say "the sky is blorple" then i actually mean that it is blue.

2 Variable, Addition

Let a variable be just a symbol that represents a literal thing.

I can define what number it represents by saying "it's equal to this"

Example:

Let

$$x = 1$$

Now if I add something to that variable, I'm going to write a double arrow "⇒" on the left of it to show that "Therefore this is true"

Example:

$$\begin{gathered} x = 1 \\ \Rightarrow x + 1 = 1 + 1 \\ \Rightarrow x + 1 = 2 \end{gathered}$$

$\Rightarrow x + 5 = 6$, as $x = 1$, we'd just put 1 in place of the x

3 Matrix

Let a "matrix" be a set of numbers ordered in a neat grid. Now we can manipulate a lot of numbers simultaneously. Interestingly enough, they are the solution to a lot of real world including machine learning and quantum mechanics.

Let's make a new <u>variable</u> called A.

Let A be a matrix of size $m \times n$ where m is the height of the matrix (how many rows it has) and n is the width (how many columns it has).

Example:

$$A = \begin{pmatrix} 1 & 2 & 5 \\ 3 & 4 & 9 \end{pmatrix}$$

See how A is also a variable? In this case, $m = 2, n = 3$ as there are 2 rows and 3 columns.

We use matrices to represent multiple numbers because they have an equivalent in the real world. In this case, quantum states.

4 Ket (State)

Let $|x\rangle$ (ket x) be a $N \times 1$ matrix where N is any positive integer (number with no decimals and is greater than 0).

Just like how x was a variable used to represent ONE number, $|x\rangle$ can be used to represent any $N \times 1$ matrix.

Example:

$$|x\rangle = \begin{pmatrix} 1 \\ 2 \\ 7 \end{pmatrix} \quad |\psi\rangle = \begin{pmatrix} 9 \\ 9 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad |\phi\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Note that ψ (psi) is pronounced sye and ϕ (phi) is pronounced fye. They are just like x but idk why quantum physicists love greek characters.

5 Photon, Wave-particle Duality

Let this be a photon. It's like a particle of light.

People, like Newton, favored the idea that light was a particle. This was until Young, a person, performed an experiment called the double slit experiment which showed light interacting like a wave. Maxwell, another person, showed that light is an electromagnetic wave — an oscillation of the electric and magnetic fields themselves. Einstein, another person, then explained the photoelectric effect, showing light also behaves as a particle, so light has both wave AND particle character.

Present day, the majority of people concur that light is a wave and particle simultaneously.

This means that photons can have a "state." These can be represented as kets!

Sometimes we observe photons having a "spin" — it's not really spinning, but physicists use the word for an intrinsic angular momentum that particles carry. Photons have spin 1; for photons this shows up as polarization (e.g. circular polarization corresponds to the two helicity states).

6 Superposition, Matrix Addition

A photon can have a probability of $x\%$ of spinning up (+). This means it would have $(100-x)\%$ of spinning down (-), since the probabilities must add up. The photon actually is both spinning up and spinning down until we observe it, where it collapses to either spinning up or spinning down. Schröndinger's cat.

We call this position of spinning both up and down simultaneously a "superposition."

Let's just say the kets $|+\rangle$ is the state of spinning up and $|-\rangle$ is the state of spinning down.

Let a "state" just be a $N \times 1$ matrix like what we just said above. So $|x\rangle, |\psi\rangle, |\phi\rangle$ are all states.

Some states can be made of other states.

Example:

Let

$$|x\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |y\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

If i multiply a matrix by a number (let this be called a "scalar"), we apply the multiplication across all values.

$$\Rightarrow 3|x\rangle = 3\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}3\times 1\\3\times 0\end{pmatrix} = \begin{pmatrix}3\\0\end{pmatrix}$$

If we add $|x\rangle + 3|y\rangle$ we get

$$\Rightarrow |x\rangle + 3|y\rangle = \begin{pmatrix}1\\0\end{pmatrix} + 3\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}1\\3\end{pmatrix}$$

Notice how we're adding the n-th elements of the states together for each row?

We can represent these two states coexisting (i.e., a superposition) using math now!

$$|\psi\rangle = \alpha|+\rangle + \beta|-\rangle$$

where α, β are just numbers. See how they are a combination of up and down states?

There is $|\alpha|^2$ probabilty that the photon, once observed, spins upwards and $|\beta|^2$ probability it spins downwards once observed.

$|a|$ is just a way of saying make a positive if it is negative.

Example:

$$|(-3)| = 3$$
$$|5| = 5$$

Note that

$$|\alpha|^2 + |\beta|^2 = 1$$

as you can't have more than 100% or less than 0% probability of something existing.

I'm going to write a ton of mathematical definitions which will build up how to do the physics in the next section!

7 Set Theory

Ok this is just for notation sake but i want to make my life a bit easier.

7.1 Element

an element is either a variable or a number or a literal in our definition

$$1, 5, a, \text{cat}, ...$$

7.2 Set

Let a set be an unordered collection of elements

Example:

This is a set S

$$S = \{3, 5, 1982549\}$$
$$S_2 = \{2n \quad | \quad n = 1, 2, 5\} = \{2, 4, 10\}$$

The | means "such that"

7.3 Subset

sets can be subsets (or equal) of another set if all elements of a set is in another set.

Example:

$$S \subset \{3, 5, 1982549, 67\}$$

We have a line underneath to show it could be a subset OR equal

$$S \subseteq \{3, 5, 1982549\}$$

7.4 In

We can say an element is "in" a set if it's in a set (duh)

Example:

$$3 \in S$$
$$4 \notin S$$

7.5 Union

The union of two sets A and B is the set of all elements that are in A or B or both.

$$A \cup B = \{x \in A \text{ or } x \in B\}$$

7.6 Intersection

The intersection of two sets A and B is the set of all elements that are in A and B.

$$A \cap B = \{x \in A \text{ and } x \in B\}$$

7.7 Standard Number System

There are some standard number systems we gotta keep in mind

$$\{1, 2, 3, ...\} \subseteq \mathbb{N}$$
$$\{..., -3, -2, -1, 0, 1, 2, 3, ...\} \subseteq \mathbb{Z}$$
$$\{\frac{p}{q} \quad \forall p \in \mathbb{Z}, q \neq 0\} \subseteq \mathbb{Q}$$
$$\{\sqrt{2}, \pi, ...\} \in \mathbb{R}$$
$$\{3 + 1i, 5 + 5i, 2 + 0i\} \in \mathbb{C} = \mathbb{F}$$

Where

- $\mathbb{N}$ are natural numbers
- $\mathbb{Z}$ are integers
- $\mathbb{R}$ are real numbers
- $\mathbb{C}$ are complex numbers, $\mathbb{F}$ are fields

where $\forall p \in \mathbb{Z}, q \neq 0$ means for all integers p AND $q \neq 0$

7.8 Partitions

A partition of a set $S = \{C_1, C_2, \ldots, C_N\}$ is a way of splitting S into

1. disjoint subsets
$$C_\alpha \cap C_\beta = \emptyset \quad \forall \quad \alpha \neq \beta$$
2. that cover S
$$\bigcup_{\alpha=1}^{N} C_\alpha = S$$

7.9 Strings

A string is an ordered sequence of symbols from a finite alphabet. The alphabet is denoted with the set notation (a set of possible symbols to use as an alphabet). Length is specified as the superscript, but if it is unlimited, then it is denoted with an asterisk $*$.

Example

$$01101011 \in \{0, 1\}^*$$
$$cba \in \{a, b, c\}^3$$
$$+ + + + - - \notin \{+, -\}^2$$

8 Function

Also let us notate a state as a function of time. What's a function?

Example:

if i let f be a function that accepts x as an argument, like this

$$f(x) = 4x$$

then

$$f(5) = 4(5) = 20$$

A function f, can also be thought of mapping numbers from a set A to set B. It's written like this

$$f : A \to B$$

A map just links a thing to another thing.

Example

Let q be a function

$$q : \mathbb{Z} \to \mathbb{Z}$$
$$q(x) = 3x$$

q essentially maps all integers to another integer, which is three times itself.

$$q(67) = 201$$

9 Summation

This is a function that adds up a series of numbers.

$$\sum_{i=1}^{n} a_i \triangleq a_1 + a_2 + \cdots + a_n$$

10 Trigonometry

There's these <u>functions</u> that are useful to know about.

$$\sin(\theta), \cos(\theta), \tan(\theta)$$

Where θ is the angle in radians.

1 radians is $\frac{180}{\pi}$ degrees. The radian to degree conversion is linear.

For the purpose of this course, we only need to think about trigonometry as functions. No need to mess around with triangles, but it sure will be helpful to know more about them.

Also know these fun facts

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$$
$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$$
$$\tan(\alpha \pm \beta) = \frac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}$$

This is called the Pythagorean identity

$$\sin^2(\theta) + \cos^2(\theta) = 1$$

11 Calculus

11.1 Limit

Let a "limit" be a function that slowly moves something towards a value. For example, if h approaches ∞, the equation below evaluates to a value. It's helpful when if we just simply set $h = 0$ or $h = \infty$, then we divide by zero or for some reason can't compute h.

$$\lim_{h \to \infty} 3 + \frac{5}{h} = 3$$

11.2 Differentiation

To differentiate something means to see how fast it changes with respect to (w.r.t) something else changing, for example, time t. Imagine you step forward in time by a very small step h. As $h \to 0$, the change in x becomes the derivative of x with respect to time.

$$\frac{d}{dt} f(t) = \lim_{h \to 0} \frac{f(t+h) - f(t)}{h}$$

To do this, us humans developed some nifty tricks. Here's all of them.

$$
\begin{aligned}
\frac{d}{dx}[f(x) \pm g(x)] &= f'(x) \pm g'(x) \\
\frac{d}{dx}[x^n] &= nx^{n-1} \\
\frac{d}{dx}[f(x)g(x)] &= f'(x)g(x) + f(x)g'(x) \\
\frac{d}{dx}[f(g(x))] &= f'(g(x))\, g'(x) \\
\frac{d}{dx}[e^x] &= e^x \\
\frac{d}{dx}[a^x] &= a^x \ln(a) \\
\frac{d}{dx}[\ln(x)] &= \frac{1}{x} \\
\frac{d}{dx}[\log_a(x)] &= \frac{1}{x \ln(a)} \\
\frac{d}{dx}[\sin(x)] &= \cos(x) \\
\frac{d}{dx}[\cos(x)] &= -\sin(x) \\
\frac{d}{dx}[\tan(x)] &= \left(\frac{1}{\cos(x)}\right)^2 \\
\frac{d}{dx}\left(\frac{1}{\tan(x)}\right) &= \frac{1}{\sin^2(x)} \\
\frac{d}{dx}\left(\frac{1}{\cos(x)}\right) &= \frac{\sin x}{\cos^2 x} \\
\frac{d}{dx}\left(\frac{1}{\sin(x)}\right) &= -\frac{\cos x}{\sin^2 x} \\
\frac{d}{dx}[\arcsin(x)] &= \frac{1}{\sqrt{1-x^2}} \\
\frac{d}{dx}[\arccos(x)] &= -\frac{1}{\sqrt{1-x^2}} \\
\frac{d}{dx}[\arctan(x)] &= \frac{1}{1+x^2} \\
\frac{d}{dx}[\operatorname{arccot}(x)] &= -\frac{1}{1+x^2} \\
\frac{d}{dx}[\operatorname{arcsec}(x)] &= \frac{1}{|x|\sqrt{x^2-1}} \\
\frac{d}{dx}[\operatorname{arccsc}(x)] &= -\frac{1}{|x|\sqrt{x^2-1}} \\
\frac{d}{dx}[f(x)^{g(x)}] &= f(x)^{g(x)} \left[g'(x)\ln(f(x)) + \frac{g(x)\, f'(x)}{f(x)}\right] \\
\frac{d}{dx}[f^{-1}(x)] &= \frac{1}{f'(f^{-1}(x))} \\
\frac{d^n}{dx^n}[f(x)g(x)] &= \sum_{k=0}^{n} \binom{n}{k} f^{(k)}(x)\, g^{(n-k)}(x)
\end{aligned}
$$

11.3 Integration

The opposite of differentiation is integration. This is the Riemann sum.

$$\int_{\alpha}^{\beta} f(t) \cdot dt = \lim_{n\to\infty} \sum_{i=1}^{n} f(t_i) \cdot (t_i - t_{i-1})$$

Integration is much more hellish. You don't need to know integration for this.

11.4 Taylor Series

Taylor, a person, once said that

$$f(x) = \sum_{n=0}^{\infty} \left(\frac{1}{n!}\right) \left(\frac{d^n f}{dx^n}\right)|_a \text{ů}(x-a)^n$$

So...

$$\begin{aligned} e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + ... \\ \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \\ \sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \end{aligned}$$

12 Complex Number

Math is complicated. Some can say it's complex.

Numbers can be complex.

Let's say we have two funny looking 2x2 matrices. Just bear with me.

$$M_R = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad M_I = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

Now we can put two scalars a, b infront of these numbers and add it up like we did before:

$$\begin{aligned} z &= a \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + b \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \\ \Rightarrow z &= \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \end{aligned}$$

We can write this weird looking matrix differently:

$$z = a + bi$$

What is i? Well, we don't know. It's not a variable! But we know what i^2 is:

$$i^2 = -1$$

Notice however much you can times a, it won't affect b, and vice versa.

12.1 Conjugate

Say

$$z = a + bi$$

Let the conjugate z^* of z be defined as this (we just make the imaginary part negative)

$$z^* = a - bi = \begin{pmatrix} a & +b \\ -b & a \end{pmatrix}$$

If you're writing in euler form, then flip the sign of the angle

$$(e^{i\theta})^* = e^{-i\theta}$$

12.2 Euler Form

Euler, another person, said that since Taylor series is a thing, we can say that

$$\begin{aligned} e^{i\theta} &= 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \cdots \\ &= 1 + i\theta - \frac{\theta^2}{2!} - i\frac{\theta^3}{3!} + \frac{\theta^4}{4!} + i\frac{\theta^5}{5!} \end{aligned}$$

If we rearrange the terms we get

$$\begin{aligned} e^{i\theta} &= \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\right) + i\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} + \cdots\right) \\ &= \cos\theta + i\sin\theta \end{aligned}$$

This means we can represent complex numbers in this weird form too:

$$z = re^{i\theta}$$

also trigometry can use complex numbers too

$$\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$$

$$\sin\theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$$

12.3 Complex Number Decomposition

This is all true

$$z = a + bi$$
$$Re(z) = \frac{1}{2}(z + z^*)$$
$$Im(z) = \frac{1}{2i}(z - z^*)$$

12.4 Absolute Value of Complex Number

$$z \cdot z^* = |z|^2$$

12.5 Complex Number Trigonometry

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$
$$\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$
$$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{1}{i} \cdot \frac{e^{i\theta} - e^{-i\theta}}{e^{i\theta} + e^{-i\theta}} = -i \cdot \frac{e^{i\theta} - e^{-i\theta}}{e^{i\theta} + e^{-i\theta}}$$

13 Linear Algebra I

More fun math with matrices

13.1 Linear Independence

Let "linearly independent states" be states that you cannot create another linearly independent state from multiplying and adding another linearly independent state.

a cleaner definition is $|\psi_1\rangle, |\psi_2\rangle, \ldots |\psi_n\rangle$ are linearly independent if the only way to get

$$c_1 |\psi_1\rangle + c_2 |\psi_2\rangle + \cdots + c_n |\psi_n\rangle = 0$$

is having $c_i = 0$ for all ($\forall$) c_i.

Example:

Given c is just a regular number,

$$|+\rangle = c\,|-\rangle \quad \text{will never happen how much you try}$$

and vice versa.

13.2 Basis State

If a set of states are BOTH

- linearly independent
- span the whole space (i.e., every state in the space can be written as a linear combination of these things)

then they are called basis states.

Note that $|+\rangle, |-\rangle$ are "basis" states

13.3 Inner product I

Inner product of states $|u\rangle$ and $|v\rangle$ is defined as

$$|u\rangle \cdot |v\rangle = u_1^* v_1 + u_2^* v_2 + ... + u_n^* v_n$$

(if the entries are real numbers, the conjugates do nothing and this reduces to the usual dot product.)

13.4 Orthogonality

State $|u\rangle, |v\rangle$ are orthogonal to each other if the inner product of state $|u\rangle$ and $|v\rangle$ are zero.

$$|u\rangle \cdot |v\rangle = 0$$

They're also known as being perpendicular!

13.5 Magnitude

Where n is the dimensions, the magnitude of $|\psi\rangle$ is defined as:

$$|| \, |\psi\rangle \, || = || \begin{pmatrix} \psi_1 \\ \psi_2 \\ \vdots \\ \psi_n \end{pmatrix} || = \sqrt{|\psi_1|^2 + |\psi_2|^2 + ... + |\psi_n|^2}$$

Note that

$$|| \, |\psi\rangle \, ||^2 = \langle \psi | \psi \rangle$$

You'll know what a bra-ket is later.

13.6 Normal

$|\psi\rangle$ is normal if it has magnitude 1

$$|| \, |\psi\rangle \, || = 1$$

13.7 Orthonormality

Let " $|+\rangle, |-\rangle$ are orthonormal" mean both $|+\rangle, |-\rangle$ are both orthogonal and normal.

13.8 Gram-Schmidt Procedure

If you have linearly independent vectors and want to convert them into an orthonormal set, you can follow the Gram-Schmidt procedure.

Given $v_1, v_2, \ldots, v_n$ as LI vectors, then $u_1, u_2, \ldots, u_n$ can be built using

$$u_k = v_k - \sum_{i=1}^{k-1} \Pi_{u_i}(v_k)$$

This makes u an orthogonal vector

Then you normalize u to get a orthonormal vector. You do this by multiplying the magnitude of the vector.

13.9 Bra, Transpose

We just learned what ket $|x\rangle$ means — a $N\times 1$ vector. Let that ALSO be called a ket. If we flip it horizontally, we get a bra $\langle x|$.

If a ket is

$$|x\rangle = \begin{pmatrix} 1+3i \\ 2+5i \\ 7 \end{pmatrix} \quad |\psi\rangle = \begin{pmatrix} 9 \\ 9 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad |\phi\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Then it's counterpart is a bra

$$\langle x| = \begin{pmatrix} 1-3i & 2-5i & 7 \end{pmatrix} \quad \langle\psi| = \begin{pmatrix} 9 & 9 & 0 & 0 & 0 \end{pmatrix} \quad \langle\phi| = \begin{pmatrix} 0 & 1 \end{pmatrix}$$

You swap the columns for the rows (transpose!) and make every element the conjugate of itself. It becomes a $1 \times N$ matrix instead.

Let this weird transformation be known as a conjugate transpose, denoted as

$$\langle v| = (|v\rangle)^{+}$$

13.10 Kronecker Delta

Where δ_{ij} is just something we define as

$$\delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

13.11 Bra-ket, matrix multiplication

Now let this be a bra-ket (get it?)

$$\langle\phi|\psi\rangle = \langle\phi|\,|\psi\rangle = \begin{bmatrix} c^* & d^* \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}$$

This means we're multiplying these two matrices $\begin{bmatrix} c^* & d^* \end{bmatrix}, \begin{bmatrix} a \\ b \end{bmatrix}$ together.

The way we multiply matrices is by performing this algorithm.

Let there be two matrices, A, B with size $m \times n$ and $n \times p$ respectively. We get a matrix $C = AB$ which is size $m \times p$.

for an element on row i, column j in C we let this be called C_{ij}

$$C_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}$$

so

$$\begin{gathered}
\langle \phi | \psi \rangle \\
= \langle \phi | \, | \psi \rangle \\
= \begin{pmatrix} c^* & d^* \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} \\
= \begin{pmatrix} c^* a + d^* b \end{pmatrix} \\
= c^* a + d^* b
\end{gathered}$$

It is a 1x1 matrix which can be simplified to a scalar value.

Note that

$$\begin{gathered}
\langle x + | x + \rangle = \langle x - | x - \rangle = 1 \\
\langle x + | x - \rangle = \langle x - | x + \rangle = 0
\end{gathered}$$

Where $x+, x-$ are orthonormal to each other. Another way we can say this is $\{|x+\rangle, |x-\rangle\}$ forms an orthonormal basis.

Let this be the notation of if/else

$$x = \begin{cases} 1 & \text{if true} \\ 0 & \text{if false} \end{cases}$$

a set of vectors $\{v_1, v_2, ..., v_n\}$ form an orthonormal basis if and only if

$$\langle v_i | v_j \rangle = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

13.12 Square Matrix

a square matrix is a matrix with the same number of rows and columns (i.e., an $N \times N$ matrix)

13.13 Symmetric Matrix

A matrix X where

$$X = X^T$$

(note that X^T means transpose)

13.14 Vector

Let vector just be the same thing as states.

13.15 Vector Space

Let a vector space V be a set of vectors with

- addition such that

$$|u\rangle + |v\rangle \in V \quad ; \quad |u\rangle, |v\rangle \in V$$

- scalar multiplication such that

13.16 Operator

Let $\hat{T}$ be an operator.

An operator is a function where

$$\hat{T} : V \to V$$

where V are the set of all possible vectors in a Hilbert space.

In quantum mechanics, all observed operators are linear which means

$$\hat{T}(a|u\rangle + b|v\rangle) = a\hat{T}(|u\rangle) + b\hat{T}(|v\rangle)$$

If we were to observe a non-linear operator, then that would cause wacky things.

also they can be represented as matrices

$$\hat{T}|\psi\rangle = \sum_m \psi'_m |m\rangle$$

where

$$\begin{bmatrix} \psi'_1 \\ \psi'_2 \\ \vdots \\ \psi'_d \end{bmatrix} = \begin{bmatrix} T_{11} & T_{12} & \cdots & T_{1d} \\ T_{21} & T_{22} & \cdots & T_{2d} \\ \vdots & & & \\ T_{d1} & T_{d2} & \cdots & T_{dd} \end{bmatrix} \begin{bmatrix} \psi_1 \\ \psi_2 \\ \vdots \\ \psi_d \end{bmatrix}$$

where $|m\rangle$ is the basis for $|\psi\rangle$

Operators must be square.

$$a|v\rangle \in V \quad ; \quad a \in \mathbb{F}, |v\rangle \in V$$

14 Hilbert Space

A Hilbert space H describes quantum space.

$$H$$

A Hilbert space is a vector space, possibly infinite-dimensional.

H has an inner product and is complete.

The states we wrote before for example $|\psi\rangle$ are states which exist in a Hilbert space.

14.1 Inner Product II

building on top of what we said about the inner product,

an inner product is any

$$\langle u|v\rangle$$

which is

1. conjugate symmetric

$$\langle u|v\rangle^* = \langle v|u\rangle$$

Example

let

$$\begin{aligned} |\psi\rangle &= \begin{bmatrix} a \\ b \end{bmatrix} \quad \Rightarrow \langle\psi| = \begin{bmatrix} a^* & b^* \end{bmatrix} \\ |\phi\rangle &= \begin{bmatrix} c \\ d \end{bmatrix} \Rightarrow \langle\phi| = \begin{bmatrix} c^* & d^* \end{bmatrix} \end{aligned}$$

so

$$\begin{aligned} (\langle\psi|\phi\rangle)^* = \left(\begin{bmatrix} a^* & b^* \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} \right)^* &= (a^*c + b^*d)^* = ac^* + bd^* = c^*a + d^*b \\ &= \langle\phi|\psi\rangle \end{aligned}$$

so ϕ, ψ are conjugate symmetric

1. linear

$$\langle u|(a|v\rangle + b|w\rangle) = a\langle u|v\rangle + b\langle u|w\rangle$$

1. positive definite

$$\forall |u\rangle \neq 0, \quad \langle u|u\rangle > 0$$

14.2 Completeness

Don't worry about it. -prof. Zhou

15 Planck-Einstein Relation

Planck, a person, first proposed (and Einstein, another person, later applied to photons via the photoelectric effect) that

$$E = hf$$

It means energy E of a photon is Planck's constant $h = 6.626 \times 10^{-34} Js$ times the frequency of the light f.

They are all variables so we can substitute them with what we want. Because it was experimental, we can't really prove with math that this is true, but we proved it with experiments. This is called the Planck–Einstein relation.

16 Angular Frequency, reduced Planck constant

Some dude in physics was like let's make a new, transformed version of frequency called angular frequency ω! (Not to be confused with angular momentum, which is a different physical quantity.)

Let

$$\omega = 2\pi f$$

Because of this, we can say that

$$\begin{aligned} E &= hf \\ &= h\frac{\omega}{2\pi} \end{aligned}$$

Let the reduced Planck constant be

$$\hbar = \frac{h}{2\pi}$$

so we can also say

$$E = \hbar\omega$$

Bohr, a person, hypothesized that atoms absorb light at discrete frequencies.

This means that atoms (photons too) have discrete (non-continuous) energy levels.

17 Particle Energy

We just said photons have discrete energy levels. Let's say they have k energy levels.

We know experimentally, that any state $|\psi(t)\rangle$ (e.g., spin) of a quantum particle can be expanded in the energy basis. Let the energy basis be just a set of basis states $E_1, E_2, \ldots, E_k$. This means a photon can be at those energies.

A state is a superposition of all those energies. Recall that a super position is just a sum ($\sum$) of those energies. I put k under the sum showing we must add all the energies and their corresponding c_k values (aka. coefficients) together.

$$|\psi(0)\rangle = \sum_k c_k |E_k\rangle$$

18 Particle Phase

We also just proved that $E = \hbar\omega$ so that particle with energy $|E_k\rangle$ will oscillate at frequency

$$\omega_k = \frac{E_k}{\hbar}$$

Oscillating means the phase of the particle changes so we can represent the phase as a complex number.

Let's say there is a two-level quantum system in some superposition. This just means

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

Because α, β can be complex, we proved experimentally that a glass polarizer, like a polarized sunglass, can change the phase of the state $|1\rangle$ for example.

Let

$$\beta' = \beta e^{i\delta}$$

Where δ can be any phase angle from 0 to 2π.

This is the same phase as which destructively interfere in the double slit experiment. That's why phases exist. Because light is a wave with a frequency (and also a particle!)

19 Schrödinger's Equation, Hamiltonian I

Recall from particle energies that

$$|\psi(0)\rangle = \sum_k c_k |E_k\rangle$$

Because we learned this thing called particle phases, we can say that particle oscillate at different phases. This is shown using phases

$$
\begin{aligned}
|\psi(t)\rangle &= \sum_k c_k e^{-i\omega_k t} |E_k\rangle \\
&= \sum_k c_k e^{-iE_k t/\hbar} |E_k\rangle
\end{aligned}
$$

If we take the derivative from both sides

$$
\begin{aligned}
\frac{d}{dt} |\psi(t)\rangle &= \frac{d}{dt} \left[\sum_k c_k e^{-iE_k t/\hbar} |E_k\rangle \right] \\
&= \sum_k c_k \cdot \left(-i \frac{E_k}{\hbar} \right) e^{-iE_k t/\hbar} |E_k\rangle \\
&= -\frac{i}{\hbar} \sum_k c_k \cdot E_k e^{-iE_k t/\hbar} |E_k\rangle
\end{aligned}
$$

Let the Hamiltonian Hbe an operator where this is true

$$
H |E_k\rangle = E_k |E_k\rangle
$$

so

$$
\begin{aligned}
&= -H \frac{i}{\hbar} \sum_k c_k \cdot e^{-iE_k t/\hbar} |E_k\rangle \\
&= -\frac{i}{\hbar} H |\psi(t)\rangle
\end{aligned}
$$

which gives us Schrödinger's equation (in state-vector form)

$$
i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle
$$

20 Quantum System

A quantum system is defined by:

1. A Hilbert space H
2. A Hamiltonian $\hat{H}$ generating time evolution via the Schrödinger equation.

The Hilbert space's dimensions defines the fixes the spin of the system.

20.1 Spin-1/2 System

This spin is for 2-dimensional Hilbert spaces. Electrons, protons, neutrons and quarks have spin-1/2. For spin-1/2 systems,

$$
S_x \triangleq \frac{\hbar}{2}\sigma_x \quad S_y \triangleq \frac{\hbar}{2}\sigma_y \quad S_z \triangleq \frac{\hbar}{2}\sigma_z
$$

i.e.,

$$S_x \triangleq \frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} \quad S_y \triangleq \frac{\hbar}{2}\begin{pmatrix}0 & -i\\ i & 0\end{pmatrix} \quad S_z \triangleq \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}$$

Note that

$$|z+\rangle = |\uparrow\rangle = |0\rangle = \begin{pmatrix}1\\ 0\end{pmatrix} \quad |z-\rangle = |\downarrow\rangle = |1\rangle = \begin{pmatrix}0\\ 1\end{pmatrix}$$

And

$$|x+\rangle = |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ 1\end{pmatrix} \quad |x-\rangle = |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ -1\end{pmatrix}$$

$$|y+\rangle = \frac{1}{\sqrt{2}}\big(|z+\rangle + i|z-\rangle\big) = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ i\end{pmatrix}, \qquad |y-\rangle = \frac{1}{\sqrt{2}}\big(|z+\rangle - i|z-\rangle\big) = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ -i\end{pmatrix}$$

20.2 Spin-1 System

This spin is for 3-dimensional Hilbert spaces. W/Z bosons, photons and gluons have spin-1. For spin-1 systems,

$$S_x \triangleq \frac{\hbar}{2}\Sigma_x \quad S_y \triangleq \frac{\hbar}{2}\Sigma_y \quad S_z \triangleq \frac{\hbar}{2}\Sigma_z$$

i.e.,

$$S_x \triangleq \frac{\hbar}{\sqrt{2}}\begin{pmatrix}0 & 1 & 0\\ 1 & 0 & 1\\ 0 & 1 & 0\end{pmatrix} \quad S_y \triangleq \frac{\hbar}{\sqrt{2}}\begin{pmatrix}0 & -i & 0\\ i & 0 & -i\\ 0 & i & 0\end{pmatrix} \quad S_z \triangleq \hbar\begin{pmatrix}1 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & -1\end{pmatrix}$$

Note that

$$|1\rangle = \begin{pmatrix}1\\ 0\\ 0\end{pmatrix} \quad |0\rangle = \begin{pmatrix}0\\ 1\\ 0\end{pmatrix} \quad |-1\rangle = \begin{pmatrix}0\\ 0\\ -1\end{pmatrix}$$

For photons, S_z is often called rectilinear and S_x is often called diagonal and S_y are often called circular.

21 Linear Algebra II

21.1 Trace

Tr(A) is only defined for a square matrix.

Given a square matrix A,

$$tr(A) = a_{11} + a_{22} + ... + a_{nn}$$

Note that

$$tr(AB) = tr(BA)$$

Traces are just adding diagonals together

$$tr(\hat{T}) = \sum_n \langle n|\,\hat{T}\,|n\rangle = \sum_n T_{nn}$$

Properties:

$$tr(\hat{A}\hat{B}) = tr(\hat{B}\hat{A})$$
$$tr(\hat{A}\hat{B}\hat{C}) = tr(\hat{B}\hat{C}\hat{A}) = tr(\hat{C}\hat{A}\hat{B})$$

Note that $tr(\hat{A}\hat{B}\hat{C}) \neq tr(\hat{A}\hat{C}\hat{B}) \rightarrow$ it must be cyclical permutation

$$tr(\alpha\hat{A} + \beta\hat{B}) = \alpha tr(\hat{A}) + \beta Tr(\hat{B})$$

Also note that if you get the traces of a 1×1 matrix, it is just

$$tr((a)) = a$$

21.2 Determinant

Det is a function that we can apply to matrices to get a scalar. It's defined for $n \times n$ only.

For a 2×2 matrix it's just

$$det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

For an $n \times n$ matrix it can be done with cofactor expansion

$$det(A) = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} M_{ij}$$

Note it's recursive so it's really expensive

also dont forget that

$$det(\hat{A}\hat{B}) = det(\hat{A})det(\hat{B})$$

An easier way to do it is through row reduction. Make a triangular matrix and multiply the diags together

$$det \begin{pmatrix} a_{11} & * & * \\ & a_{22} & * \\ 0 & 0 & a_{33} \end{pmatrix} = a_{11}a_{22}a_{33}$$

21.3 Matrix Invertibility

<u>Operator</u> $\hat{T}$ (i.e., a matrix) is invertible if and only if $det(\hat{T}) \neq 0$

21.4 Eigenvector/Eigenvalue

https://www.youtube.com/watch?v=PFDu9oVAE-g

$$A\vec{v} = \lambda\vec{v}$$

Basically if I apply matrix transformation A to a vector $|v\rangle$ and it still points in the same direction, it is an Eigenvector. It could be multiplied by a scalar value λ (Eigenvalue) after applying the transform, which means its magnitude need not be preserved by the transform.

The set of all vector $|v\rangle$'s are Eigenvectors of transformation A.

An Eigenstate are states instead of vectors, operators instead of matrices but are the same thing.

21.5 Eigenstates

Eigenstates are eigenvectors of operators.

$$\hat{A}\,|\psi\rangle = \lambda\,|\psi\rangle$$

21.6 Eigenspace

An eigenspace of operator $\hat{A}$ is the set of all eigenvectors/eigenstates with the same eigenvalue plus the zero vector.

Example

If operator $\hat{A}$ has eigenvalue λ, then the λ-eigenspace of the operator $\hat{A}$ is

$$\{|\psi\rangle : \hat{A}\,|\psi\rangle = \lambda\,|\psi\rangle\}$$

21.7 Eigenbasis

An eigenbasis of operator $\hat{A}$ is a set of eigenvectors/eigenstates that form a basis for the Hilbert space.

Every Hermitian operator has an eigenbasis.

21.8 Characteristic Polynomial

polynomial $p(\lambda) = \alpha_0\lambda^n + \alpha_1\lambda^{n-1} + ... + \alpha_{n-1}\lambda^1 + \alpha_n$

$$p(\lambda) = det(A - \lambda I)$$

Solving

$$p(\lambda) = 0$$

Gives λ values which are eigenvalues of A

21.9 Subspace

A subset $U \subseteq V$ is a subspace if it satisfies:

1. contains zero vector
2. closed under addition $|u\rangle, |v\rangle \in U, |u\rangle + |v\rangle \in U$
3. closed under scalar multiplication $|u\rangle \in U, c \in \mathbb{F} \Rightarrow c|u\rangle \in U$

21.10 Span Function

A span of vectors is the set of all linear combinations you can build from them

21.11 Invariant Subspace

A subspace $U \subseteq V$ is called $\hat{T}$-invariant if operator $\hat{T}$ keeps everything in U inside U. If it is T-invariant, then this must always be true:

$$\hat{T}(U) = \{\hat{T}|\psi\rangle \,|\, |\psi\rangle \in U\} \subseteq U$$

Simplest T invariant subspace is (with dimensions d)

$$U = span\{|u\rangle\}$$
$$\hat{T}|y\rangle = \lambda|u\rangle$$

When λ is Eigenvalue and $|u\rangle$ is eigenvector and $|u\rangle \neq 0$ and you can define these vectors by checking

$$(\hat{T} - \lambda I)|u\rangle = 0$$
$$0 = det(\hat{T} - \lambda I)$$
$$= (\lambda_1 - \lambda)(\lambda_2 - \lambda)..(\lambda_d - \lambda)$$

In general, this determinant has degree λ_i are zeroes of character polynomial $\lambda_i \in C$ can be repeated

21.12 Diagonalization

A matrix is diagonalizable if it has linearly independent eigenvectors which occurs if the geometric multiplicity (dimension of the eigenspace) equals the algebraic multiplicity (root multiplicity in the characterstic polynomial) for every eigenvalue.

If matrix A is diagonalizable then it can be written as

$$A = PDP^{-1}$$

Where $D = diag(\lambda_1, \ldots, \lambda_n)$

21.13 Matrix Exponential

The matrix exponential of square matrix X is defined by a power series

$$e^X \triangleq \sum_{n=0}^{\infty} \frac{X^n}{n!}$$

Properties of this

- differential equations
  - the function $U(t) = e^{tX}$ is the only unique soln. to

$$\frac{dU}{dt} = XU(t) \quad U(0) = I$$

- diagonalizable matrices → if X is diagonalizable (which is guaranteed for, e.g., real symmetric or Hermitian matrices) then we can write $X = PDP^{-1}$ with $D = diag(\lambda_1, \dots \lambda_n)$ then

$$e^X = Pdiag(e^{\lambda_1}, ..., e^{\lambda_n})P^{-1}$$

where P is the basis matrix — a matrix whose columns are the eigenvectors X w.r.t $\hat{A}$.

Specifically, if $v_1, v_2, \dots, v_n$ are eigenvectors of X corresponding to eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ then

$$P = \begin{bmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{bmatrix}$$

- Hermitian and skew-Hermitian matrices → if $H = H^+$ then e^{iH} is unitary this means

$$(e^{iH})^+ e^{iH} = I$$

Note that if A is a matrix, then, via Taylor series, we have

$$e^{At} = \sum_{k=0}^{\infty} \frac{A^k t^k}{k!} = I + At + \frac{(At)^2}{2!} + \frac{(At)^3}{3!} + \cdots$$

$$\frac{d}{dt} e^{At} = \sum_{k=1}^{\infty} \frac{kA^k t^{k-1}}{k!} = A \sum_{k=1}^{\infty} \frac{A^{k-1} t^{k-1}}{(k-1)!} = Ae^{At} = e^{At}A$$

21.14 Commutator

The commutator of two operators $\hat{A}, \hat{B}$ is defined as

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

If $[\hat{A}, \hat{B}] = 0$ then that means you can commute matrix multiplication.

If "$\hat{A}$ commutes with $\hat{B}$", then that means $\hat{A}\hat{B} = \hat{B}\hat{A}$ which also means $[\hat{A}, \hat{B}] = 0$.

21.15 Adjoint

An adjoint is basically a conjugate transpose but for an operator (whereas a conjugate transpose can only be applied to a matrix)

Given operator $\hat{A}$, it's adjoint is the operator $\hat{A}^+$ such that $\forall \left|i\right\rangle, \left|j\right\rangle$

$$\left\langle i\right| \hat{A}^+ \left|j\right\rangle = (\left\langle j\right| \hat{A} \left|i\right\rangle)^*$$
$$(\hat{A}^+)_{ij} = \hat{A}^*_{ji}$$

Where, for any operator $\hat{O}$, a matrix element $\left\langle i\right| \hat{O} \left|j\right\rangle$ is just the (i, j)-th element of the matrix representing $\hat{O}$ in basis $\{\left|i\right\rangle\}$

Note that $\left|i\right\rangle$ and $\left|j\right\rangle$ are arbitrary basis states

where A^* is the complex conjugation of A

properties:

$$(\hat{A}^+)^+ = \hat{A}$$
$$(\hat{A} + \hat{B})^+ = \hat{A}^+ + \hat{B}^+$$
$$(\hat{A}\hat{B})^+ = \hat{B}^+\hat{A}^+$$

Example:

Consider operator

$$\hat{A} = \left|1\right\rangle \left\langle 2\right| + 2i \left|2\right\rangle \left\langle 1\right| + 3 \left|2\right\rangle \left\langle 2\right|$$
$$\Rightarrow \text{adjoint of } \hat{A} = \left|2\right\rangle \left\langle 1\right| - 2i \left|1\right\rangle \left\langle 2\right| + 3 \left|2\right\rangle \left\langle 2\right|$$

So w.r.t. to basis $\{\left|1\right\rangle, \left|2\right\rangle\}$, $\hat{A}$ is

$$\hat{A} = \begin{pmatrix} 0 & 1 \\ 2i & 3 \end{pmatrix}$$

The conjugate transpose of $\hat{A} = \hat{A}^+$ which is

$$\hat{A}^+ = \begin{pmatrix} 0 & -2i \\ 1 & 3 \end{pmatrix}$$

Which, once we pick an orthonormal basis, the matrix of $\hat{A}^+$ equals the conjugate transpose of the matrix of $\hat{A}$

$$\hat{A}^+ = \text{adjoint of } \hat{A}$$

21.16 Normal Operator

An operator A is normal if and only if

$$\hat{A}\hat{A}^{+} = \hat{A}^{+}\hat{A}$$

Not to be confused with normal.

21.17 Hermitian Matrix

Let a "Hermitian matrix" be a matrix that equals its own conjugate transpose

If and only if A is a matrix and is Hermitian, then this is true

$$A = A^{+}$$

This means a bra and ket are Hermitian matrices of each other!

Another example:

$$|\psi\rangle = \begin{bmatrix} a \\ b \end{bmatrix} \quad |\phi\rangle = \begin{bmatrix} c \\ d \end{bmatrix} \Rightarrow \langle\phi| = \begin{bmatrix} c^* & d^* \end{bmatrix}$$

21.18 Hermitian Operator

An operator is Hermitian if and only if the adjoint of the operator is equal to itself.

Properties

- all eigenvalues of a Hermitian operator are real
- All eigenvectors of a Hermitian operator form an orthonormal basis
- observables are usually some kind of Hermitian operator

The eigenvalues represent possible measurement outcomes. Orthonormal eigenstates $|\alpha\rangle$ mean that these are definite value states

21.19 Anti-Hermitian Operator

An operator is anti-hermitian if and only if

$$\hat{B}^{+} = -\hat{B}$$

note that if you do

$$\hat{A} = i\hat{B}$$

then $\hat{A}$ is Hermitian

Any operator $\hat{T}$ can be written as linear combination of Hermitian and anti-Hermitian operator.

$$\hat{T} = \hat{A} + i\hat{B}$$
$$\hat{A} = \frac{\hat{T} + \hat{T}^+}{2}$$
$$\hat{B} = \frac{\hat{T} - \hat{T}^+}{2i}$$

Note that this is true for regular complex numbers

21.20 Positive Semi-Definite (PSD) Operator

An operator $\hat{P}$ is PSD if and only if

- $\hat{P}$ is Hermitian
- this holds true for all states $|\psi\rangle$:

$$\langle\psi|\,\hat{P}\,|\psi\rangle \geq 0$$

This means the angle between $|\psi\rangle$ and $\hat{P}\,|\psi\rangle$ is less than 90 degrees.

Example

- Projectors are PSD

21.21 Projection

The projection of v onto a vector u is

$$\Pi_u(v) = \frac{\langle v|u\rangle}{\langle u|u\rangle}u$$

Example

$$|\psi\rangle = \begin{bmatrix} a \\ b \end{bmatrix} \Rightarrow \langle\psi| = \begin{bmatrix} a^* & b^* \end{bmatrix}$$
$$|\phi\rangle = \begin{bmatrix} c \\ d \end{bmatrix} \Rightarrow \langle\phi| = \begin{bmatrix} c^* & d^* \end{bmatrix}$$

$$\Pi_\psi(\phi) = \frac{\begin{bmatrix} c^* & d^* \end{bmatrix}\begin{bmatrix} a \\ b \end{bmatrix}}{\begin{bmatrix} a^* & b^* \end{bmatrix}\begin{bmatrix} a \\ b \end{bmatrix}}\begin{bmatrix} a \\ b \end{bmatrix}$$
$$= \frac{c^*a + d^*b}{a^*a + b^*b}\begin{bmatrix} a \\ b \end{bmatrix}$$

This is similar to the dot product, but it gives a vector

21.22 Projector

Projectors are operators with a special property

$$\hat{\Pi}^2 = \hat{\Pi}$$

A projector operator (aka. a projector) is a projection that

$$\hat{\Pi}_u = \frac{|u\rangle \langle u|}{\langle u|u\rangle}$$

Usually, by convention, we pick the representative state $|u\rangle$ to be normalized, i.e., $\langle u|u\rangle = 1$. So sometimes it can be written as

$$\hat{\Pi}_u = |u\rangle \langle u| \quad \text{if} \quad \langle u|u\rangle = 1$$

once i apply it to a state v, i get the projection of that state v on state u

21.23 Unitary Operator

An operator is unitary if and only if

$$\hat{U}^+\hat{U} = \hat{U}\hat{U}^+ = I$$

- it preserves inner product

$$\begin{aligned} |\alpha'\rangle &= \hat{U}\,|\alpha\rangle \\ |\beta'\rangle &= \hat{U}\,|\beta\rangle \\ \langle\alpha'|\beta'\rangle &= \langle\alpha|\,\hat{U}^+\hat{U}\,|\beta\rangle \\ &= \langle\alpha|\,I\,|\beta\rangle \\ &= \langle\alpha|\beta\rangle \end{aligned}$$

- U is basis chagne between orthonormal basis because you can write $\hat{U}$ as
  - as chatgpt puts it: unitary matrices/operators represent a rotation or change of coordinates in Hilbert space, without changing lengths or angles.

$$\hat{U} = \hat{U} I$$

I can use completeness relation so that if

$$\hat{U}\,|\alpha_n\rangle = |\beta_n\rangle$$

then we can change basis

$$
\begin{aligned}
&\hat{U}\sum_n |\alpha_n\rangle\langle\alpha_n| \\
&= \sum_n U|\alpha_n\rangle\langle\alpha_n| \\
&= \sum_n |\beta_n\rangle\langle\alpha_n|
\end{aligned}
$$

Both $\{|\alpha_n\rangle\}$ and $\{|\beta_n\rangle\}$ are orthonormal basis

21.24 Spectral Decomposition

Let there be a Hermitian operator $\hat{A}$ on a finite-dimensional Hilbert space. Spectral theorem says that a $\hat{A}$ is unitarily diagonalizable and only if it is normal. There exists real eigenvalues $\{a_n\}$, orthonormal eigenvectors $\{|k\rangle\}$ which form orthogonal projectors $\{\hat{\Pi}_n\}$ such that

$$\hat{A} = \sum_{n=1}^{N} a_n \hat{\Pi}_n = \sum_{n=1}^{N} a_n |k\rangle\langle k|$$

where projectors satisfy

1. $\hat{\Pi}_n\hat{\Pi}_m = 0$ for $n \neq m$ which means they are orthogonal
2. $\sum_{n=1}^{N} \hat{\Pi}_n = \hat{I}$ which means they are a complete set of projectors
3. this projector projects onto the a_n-eigenspace of $\hat{A}$

So $\{a_n\}$ is the spectrum of $\hat{A}$.

And that means you can think $\hat{A}$ as scaling each eigenspace by its own eigenvalue.

22 Observable

An observable is represented by a Hermitian operator acting on a Hilbert space.

23 Pauli Matrices

Three matrices which are defined as

$$
\begin{aligned}
\sigma^0 = \sigma_0 = I_2 &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\
\hat{\sigma}_x = \sigma^1 = \sigma_1 &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\
\hat{\sigma}_y = \sigma^2 = \sigma_2 &= \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \\
\hat{\sigma}_z = \sigma^3 = \sigma_3 &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}
\end{aligned}
$$

Properties

- spacing

$$\hat{\sigma}_i^2 = I_2 \quad \forall \quad i \in \{x, y, z\}$$

- commutation

note that

The symbol ε_{ijk} is the Levi-Civita symbol, an antisymmetric tensor that encodes signs based on the order of its indices.

- in 3D

$$\varepsilon_{ijk} = \begin{cases} +1 & \text{if } (i,j,k) \text{ is an even permutation of (1,2,3)} \\ -1 & \text{if } (i,j,k) \text{ is an odd permutation of (1,2,3)} \\ 0 & \text{if any index repeats} \end{cases}$$

- in 2D (only two indices, written ε_{ij})

$$\varepsilon_{ij} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$[\sigma_i, \sigma_j] = 2i\varepsilon_{ijk}\sigma_k$$

where an "even" permutation is (1,2,3), (2,3,1), (3,1,2) and an "odd" is (2,1,3), (1,3,2), (3,2,1)

- anti commutation

Note that the anticommutator is

$$\{A, B\} = AB + BA$$

$$\{\sigma_i, \sigma_j\} = 2\delta_{ij}I_2$$

- completeness

$$\sigma_0 = I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

the set $\{\sigma_0, \sigma_x, \sigma_y, \sigma_z\}$ spans all 2x2 complex matrices

$$\{I, X, Y, Z\} \quad \text{form a basiss for operator } L(\mathbb{C})^2$$

where

forms a basis for all 2x2 complex matrices. Any 2x2 matrix

$$M = \sum_{\mu=0}^{3} c_\mu \sigma_\mu$$

- whatever this is

$$
\begin{gathered}
\vec{\sigma} = (X, Y, Z) \quad \text{are Hermitian} \\
\sigma^1 = \sigma^x = X \\
\sigma^2 = \sigma^y = Y \\
\sigma^3 = \sigma^z = Z \\
X^2 + Y^2 + Z^2 = I \\
tr(X) = tr(Y) = tr(Z) = 0 \\
XY = iZ, YZ = iX, ZX = iY \\
[X, Y] = i\hbar Z, \quad [Y, Z] = i\hbar X, \quad [Z, X] = i\hbar Y \\
YX = (XY)^+ = (iZ)^+ = -iZ
\end{gathered}
$$

and

$$\hat{A} = a_0\sigma_0 + a_1\sigma_x + a_2\sigma_y + a_3\sigma_z \quad a_i \in \mathbb{C}$$

$\hat{A} \cdot \hat{B}$ can be expanded as

$$\hat{A}\hat{B} = \sum_{i=0}^{3} c_i \sigma^i$$

If $c_i \in \mathbb{R}$, then $\hat{A}$ is <u>Hermitian</u>

24 Bloch Sphere

A general pure state of a qubit is

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle \quad \theta \in [0, \pi] \quad \phi \in [0, 2\pi]$$

Where the orthogonal states are antipodal (North pole is $|0\rangle$ and South pole is $|1\rangle$). The equator contains equal superpositions like $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$.

A state can be on the inside of a Bloch sphere which means its a mixed state. If a state is on the surface of the Bloch sphere, it is a pure state.

A rotation by angle θ around the axis $\hat{n}$ is given by

$$R_{\hat{n}}(\theta) = e^{-i\theta\,\hat{n}\cdot\vec{\sigma}/2} = \cos\left(\tfrac{\theta}{2}\right) I - i\sin\left(\tfrac{\theta}{2}\right)(n_x\sigma_x + n_y\sigma_y + n_z\sigma_z)$$

Where

$$
\begin{gathered}
R_x(\theta) = \begin{pmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix} \\
R_y(\theta) = \begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix} \\
R_z(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}
\end{gathered}
$$

25 Observable on a qubit

Any observable on a qubit can be written as

$$\hat{A} = a_0\sigma_0 + \vec{a}\vec{\sigma_{1,2,3}}$$

You can think of $a_0\sigma_0$ as the scalar value and $\vec{a}\vec{\sigma_{1,2,3}}$ as it helps with the Bloch Sphere.

This can be written in spherical units too

$$\hat{n} = (n_x, n_y, n_z) = (\sin\theta\cos\psi, \sin\theta\sin\psi, \cos\theta)$$

spin along $\hat{n}$ can be described as $\hat{S} = \frac{\hbar}{2}\vec{\sigma}$. This is known as the "spin along the unit vector $\hat{n}$". To do this we project $\hat{S}$ onto $\hat{n}$.

$$\begin{aligned}
\hat{S}_n = \hat{S}\cdot\hat{n} &= \frac{\hbar}{2}\vec{\sigma}\cdot\hat{n} \\
&= \frac{\hbar}{2}\left(n_x\sigma_x + n_y\sigma_y + n_z\sigma_z\right) \\
&= n_x\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} + n_y\begin{pmatrix}0 & -i\\ i & 0\end{pmatrix} + n_z\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix} \\
&= \begin{pmatrix} n_z & n_x - in_y \\ n_x + in_y & -n_z\end{pmatrix} \\
\Rightarrow \hat{S}_{\hat{n}} &= \frac{\hbar}{2}\begin{pmatrix} n_z & n_x - in_y \\ n_x + in_y & -n_z\end{pmatrix}.
\end{aligned}$$

We work out the eigenvalues by doing

$$\begin{aligned}
det(\hat{S}_{\hat{n}} - \lambda I) &= 0 \\
\Rightarrow \left(\tfrac{\hbar}{2}n_z - \lambda\right)\left(-\tfrac{\hbar}{2}n_z - \lambda\right) - \left(\tfrac{\hbar}{2}\right)^2(n_x - in_y)(n_x + in_y) &= 0 \\
\Rightarrow \lambda^2 - \left(\tfrac{\hbar}{2}\right)^2(n_x^2 + n_y^2 + n_z^2) &= 0 \\
\lambda^2 = \left(\tfrac{\hbar}{2}\right)^2|\hat{n}|^2 &\Rightarrow \lambda = \pm\tfrac{\hbar}{2}|\hat{n}|
\end{aligned}$$

Because $\hat{n}$ is a unit vector,

$$\lambda = \pm\tfrac{\hbar}{2}$$

26 Degeneracy

Two or more linear independent quantum states (eigenstates) share the same eigenvalue. An energy level E_n is degenerate if there exist multiple linearly independent eigenstates $|\psi_1\rangle, |\psi_2\rangle, \dots, |\psi_n\rangle$ which satisfy

$$\hat{H}|\psi_i\rangle = E_n|\psi_i\rangle \quad \forall i = 1, 2, \dots, g$$

Where if $g = 1$, the energy level is non-degenerate. If $g > 1$, the energy level is degenerate. If something is degenerate, that implies there is no unique basis. What you want to write down should be a projection onto the subspaces.

$$\hat{H} \triangleq \begin{pmatrix} E_0 & & 0 \\ & E_1 & \\ 0 & & E_2 \end{pmatrix} \quad |0\rangle \triangleq \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad |1a\rangle \triangleq \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad |1b\rangle \triangleq \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\Rightarrow \hat{H} = E_0 |0\rangle \langle 0| + E_1 |1a\rangle \langle 1a| + E_2 |1b\rangle \langle 1b|$$

The states $|1a\rangle, |1b\rangle$ are degenerate if and only if $E_1 = E_2$. If $E_0 = E_1, E_1 \neq E_2$ then $|0\rangle$ is degenerate with $|1a\rangle$.

This just means that you can't detect the difference between states given only the energy/eigenvalue.

Remember the eigenvalue is observed as energy if the operator is the Hamiltonian.

This means the Hamiltonian alone is not enough to determine the state.

This is because if E is degenerate then

$$H |\psi_1\rangle = E |\psi_1\rangle$$
$$H |\psi_2\rangle = E |\psi_2\rangle$$

Now just finding E doesn't mean we can distinguish the states.

We need another operator $\hat{A}$ which commutes with $\hat{H}$ to distinguish the states. Note that just because $\hat{A}$ commutes with $\hat{H}$ doesn't mean we can distinguish the states.

$$A |\psi_1\rangle = a_1 |\psi_1\rangle$$
$$A |\psi_2\rangle = a_2 |\psi_2\rangle$$

And if $a_1 \neq a_2$ then we can distinguish the states.

BUT if $[\hat{A}, \hat{H}] \neq 0$ then this doesn't always hold true for the entire Hilbert Space.

This is because if

$$\hat{H} |\psi\rangle = E |\psi\rangle$$
$$\Rightarrow \hat{A}\hat{H} |\psi\rangle = \hat{A}E |\psi\rangle$$
$$\Rightarrow (\hat{H}\hat{A} - [\hat{H}, \hat{A}]) |\psi\rangle = E\hat{A} |\psi\rangle$$
$$\Rightarrow \hat{H}\hat{A} |\psi\rangle - [\hat{H}, \hat{A}] |\psi\rangle = E\hat{A} |\psi\rangle$$

If $[\hat{H}, \hat{A}] = 0$ (i.e., H and A commute) then

$$\Rightarrow \hat{H}\hat{A} |\psi\rangle = E\hat{A} |\psi\rangle$$

This means that $\hat{A} |\psi\rangle$ is in the E-eigenspace of $\hat{H}$. Another way to say this is that $\hat{A} |\psi\rangle$ is in the set of eigenvectors for H that share eigenvalue E.

This means that $\hat{A}$ maps a vector in the E-eigenspace of $\hat{H}$ to another vector in the E-eigenspace of $\hat{H}$.

27 Compatibility

Compatibility is a property of two operators.

If $|\phi_n\rangle$ is the n-th eigenstate of $\hat{A}$ (i.e., the eigenbasis of $\hat{A}$),

Let

$$\hat{A} = \sum_{n=1}^{N} a_n |\phi_n\rangle \langle\phi_n|$$

If $\hat{B}$ is compatible with $\hat{A}$ (i.e., $\hat{B}$ is diagonalizable with $\hat{A}$) then,

$$\hat{B} = \sum_{n=1}^{N} b_n |\phi_n\rangle \langle\phi_n|$$

27.1 Compatibility of Observables

If two observables are compatible, then they can be measured simultaneously with one basic measurement. This means that observing one doesn't affect the other.

Two operators $\hat{A}$ and $\hat{B}$ are compatible if and only if they commute, i.e. $[\hat{A}, \hat{B}] = 0$.

$$\hat{A}, \hat{B} \quad \text{are compatible} \quad \Leftrightarrow \quad [\hat{A}, \hat{B}] = 0$$

Proof

$$\Rightarrow \hat{A}\hat{B} = \sum_{n,m} A_n B_m |\phi_n\rangle \langle\phi_n|\phi_m\rangle \langle\phi_m|$$

$$= \sum_{n} A_n B_n |\phi_n\rangle \langle\phi_n| = \hat{B}\hat{A}$$

$$\Leftarrow \hat{A} |\phi_n\rangle = A_n |\phi_n\rangle$$

$$\hat{A}\hat{B} |\phi_n\rangle = \hat{B}\hat{A} |\phi_n\rangle = A_n \hat{B} |\phi_n\rangle$$

Example

28 Probability Theory

probability space Σ has points and elements and associated probabilities with them

example:

coin flip

$$\sum = \{e : H, T\} \rightarrow \frac{1}{2}$$

Sum of all probability of your space should be 1

$$\sum P(\Sigma) = 1$$

We have a random variable X. You have an event in that random variable space called x. The probability of finding that event in that random variable distribution

$$P(X) = \sum_{e \in V_x} P(e)$$

where

$$V_x = \{e : X(e) = x\}$$

Note that the probabilities are always between 0 and 1

$$0 \leq P(x) \leq 1$$

28.1 Joint Random Variable

We have a joint random variable (X, Y) and we want to find a pair of values

$$P(x, y) = \sum_{x,y \in V_{x,y}} P(e)$$
$$V_{x,y} = \{e : X(e) = x, Y(e) = y\}$$

If i have one variable, then the other one can be inferred from the other.

$$P(X) = \sum_{y} P(x, y)$$
$$P(Y) = \sum_{x} P(x, y)$$

28.2 Conditional Probability

from a joint distribution $\rightarrow$ if I select y for Y with $P(Y) \neq 0$

What is the probability x given y

$$P(X|Y) = \frac{P(X, Y)}{P(Y)}$$

Special case

If X, Y are independent (if X occurs, Y has no effect, and vice versa)

$$P(X, Y) = P(X)P(Y)$$

28.3 Numerical Random Variable/Expected Value

Uncertainty of a certain measurement

- a numerical RV X labels events by values that are real numbers X.
- the mean or expected value

$$\mathbb{E}[X] = \langle X \rangle = \sum_X X \cdot P(x)$$

28.4 Spread

How do we measure spread from $\mathbb{E}[X]$?

Let $\mu = \mathbb{E}[X]$

$$\begin{aligned} Var(X) &= \left\langle (x-\mu)^2 \right\rangle = \left\langle X^2 \right\rangle - 2\mu \langle X \rangle + \mu^2 \\ &= \left\langle X^2 \right\rangle - \langle X \rangle^2 \end{aligned}$$

28.5 Standard Deviation

it's the square root of the variance

$$\Delta X = \sqrt{Var(X)} = \sqrt{\mathbb{E}[X^2] - \mathbb{E}[X]^2}$$

28.6 Expected Value of an Observable

To determine the mean (EV) of an <u>operator</u> $\hat{A}$, we must know the state $|\psi\rangle$ you're in.

$$\begin{aligned} \mathbb{E}(A) &= \sum_n A_n P(n) \\ &= \sum A_n |\langle n|\psi\rangle|^2 \\ &= \sum A_n \langle \psi|n\rangle \langle n|\psi\rangle \\ &= \langle \psi| \sum_n A_n |n\rangle \langle n| |\psi\rangle \\ &= \langle \psi| \hat{A} |\psi\rangle \\ &\triangleq \left\langle \hat{A} \right\rangle \end{aligned}$$

Another convientn reprsentation is

$$\begin{aligned} \hat{\rho} &\triangleq |\psi\rangle \langle\psi| \\ \Rightarrow \left\langle \hat{A} \right\rangle &= tr(\hat{\rho}\hat{A}) \\ &= tr(|\psi\rangle \langle\psi| \hat{A}) \\ & tr(\langle\psi| \hat{A} |\psi\rangle) \end{aligned}$$

where tr is <u>trace</u>

28.7 Cauchy-Schwarz Inequality

Given two vectors $|u\rangle, |v\rangle \in \mathbb{C}^n$,

$$|\langle u|v\rangle| \leq \|u\|\|v\|$$

It's squared form can be written as

$$|\langle u|v\rangle|^2 \leq \langle u|u\rangle \langle v|v\rangle$$

29 Born's Rule

If a quantum system is in state $|\psi\rangle$ and you measure observable $\hat{A}$ — the probability of getting eigenvalue λ_α (which corresponds to eigenstate $|\alpha\rangle$) is

$$P_\alpha = |\langle \alpha|\psi\rangle|^2$$

This only holds true if eigenstate $|\phi_n\rangle$ is non-degenerate and normalized. Usually, the convention is that we pick the representative state $\langle \phi_n|\phi_n\rangle = 1$ (i.e., it is normalized).

Born's rule is a postulate, meaning that it is based on experimental observations and cannot be derived from first principles.

30 Measuring a Quantum State

Before: The system is in a state $|\psi\rangle$ and you measure the observable $\hat{A}$.

Measure: We get an outcome A_n with probability $P_n = \langle\psi| \hat{\Pi}_n |\psi\rangle$

Given Born's rule for a non-degenerate eigenstate $|\phi_n\rangle$,

$$P_n = |\langle \phi_n|\psi\rangle|^2$$

This is because Born's rule. Note that $\hat{\Pi}_n$, the projector onto the λ_n-eigenspace of $\hat{A}$, is

$$\hat{\Pi}_n = |\phi_n\rangle \langle \phi_n|$$

$$\Rightarrow \hat{\Pi}_n |\psi\rangle = |\phi_n\rangle \langle \phi_n|\psi\rangle$$

$$\Rightarrow = \langle \phi_n|\psi\rangle |\phi_n\rangle$$

So

$$\Rightarrow ||\hat{\Pi}_n |\psi\rangle ||^2$$

$$\Rightarrow = || \langle \phi_n|\psi\rangle |\phi_n\rangle ||^2$$

$$\Rightarrow = |\langle \phi_n|\psi\rangle|^2 || |\phi_n\rangle ||^2$$

We know that the magnitude of the eigenstate $|\phi_n\rangle$ is 1.

After: State collapses to

$$|\psi'\rangle = \frac{1}{\sqrt{P_n}} \hat{\Pi}_n |\psi\rangle$$

.

The collapse is true only if this is an idealized measurement that is non-destructive. This means that the measurement must not destroy the system (e.g., photon absorbed by detector). There's no noise as well.

31 Density Matrix

Let ρ be a matrix.

ρ is a density matrix if it is:

- Hermitian
- Positive semi-definite
- Trace $Tr(\rho) = 1$

It contains all information needed to predict measurement outcomes. It is fundementally the probability distribution over quantum states. For the observable A we see that the expected value is

$$\mathbb{E}[A] = tr(\rho A)$$

32 Heisenberg Uncertainty Relation

$$\Delta A \Delta B \geq \frac{1}{2} \left| \mathbb{E}\left[[\hat{A}, \hat{B}] \right] \right|$$

$$\Delta A^2 \Delta B^2 \geq \left| \langle \psi | \frac{1}{2j} [\hat{A}, \hat{B}] | \psi \rangle \right|^2 = \left(\frac{1}{2j} \mathbb{E}\left[[\hat{A}, \hat{B}] \right] \right)^2$$

Proof Consider any scalar $\lambda \in \mathbb{C}$ and any vectors $|u\rangle, |v\rangle \in \mathbb{C}^n$ and we have two states $|f\rangle$ and $|g\rangle$.

We subtract it like so $|f\rangle - \lambda |g\rangle$ is also a vector

Note that for QM systems, $\langle u|u\rangle = 1$. This is because of Born's rule where the number it represents is the total probability, which needs to sum to 1.

$$\langle f - \lambda g | f - \lambda g \rangle \geq 0$$

$$\Rightarrow \langle f|f\rangle - \lambda \langle f|g\rangle - \lambda^* \langle g|f\rangle + (|\lambda|)^2 \langle g|g\rangle \geq 0$$

We want to choose the λ that minimizes the expression.

$$\lambda = \frac{\langle g|f\rangle}{\langle g|g\rangle} \quad \Rightarrow \quad \lambda^* = \frac{\langle f|g\rangle}{\langle g|g\rangle}, \quad |\lambda|^2 = \frac{\langle f|g\rangle \langle g|f\rangle}{\langle g|g\rangle^2}$$

Which makes $|f\rangle - \lambda |g\rangle$ orthogonal to $|g\rangle$.

$$\Rightarrow \langle f|f\rangle - \frac{\langle g|f\rangle}{\langle g|g\rangle} \langle f|g\rangle - \frac{\langle f|g\rangle}{\langle g|g\rangle} \langle g|f\rangle + \frac{\langle f|g\rangle \langle g|f\rangle}{\langle g|g\rangle^2} \langle g|g\rangle \geq 0$$

Note that $\langle g|f\rangle \langle f|g\rangle = |\langle f|g\rangle|^2$

$$\Rightarrow \langle f|f\rangle - \frac{|\langle f|g\rangle|^2}{\langle g|g\rangle} - \frac{|\langle f|g\rangle|^2}{\langle g|g\rangle} + \frac{|\langle f|g\rangle|^2}{\langle g|g\rangle} \geq 0$$

$$\Rightarrow \langle f|f\rangle - \frac{|\langle f|g\rangle|^2}{\langle g|g\rangle} \geq 0$$

Assuming $\langle g|g\rangle \geq 0$

$$\Rightarrow \langle f|f\rangle \langle g|g\rangle \geq |\langle f|g\rangle|^2$$

This is Cauchy-Schwarz squared

Consider the observables that we defined above. This can be written as

$$|f\rangle = (\hat{A} - \mathbb{E}[A]I) |\psi\rangle \quad |g\rangle = (\hat{B} - \mathbb{E}[B]I) |\psi\rangle$$

Note that ΔA is the standard deviation of $\hat{A}$.

$$\langle f|f\rangle = \langle\psi|\,(\hat{A} - \mathbb{E}[A]I)\,|\psi\rangle$$
$$= \langle\psi|\left(\hat{A}^2 - 2\mathbb{E}[A]\hat{A} + \mathbb{E}[A]^2 I\right)|\psi\rangle$$
$$= \langle\psi|\,\hat{A}^2\,|\psi\rangle - 2\mathbb{E}[A]\,\langle\psi|\,\hat{A}\,|\psi\rangle + \mathbb{E}[A]^2\,\langle\psi|\,I\,|\psi\rangle$$
$$= \mathbb{E}[\hat{A}^2] - 2\mathbb{E}[A]^2 + \mathbb{E}[A]^2$$
$$= \mathbb{E}[\hat{A}^2] - \mathbb{E}[A]^2$$
$$= \Delta A^2$$

By symmetry

$$\langle g|g\rangle = \Delta B^2$$

Plug into our equation above

$$\Delta A^2 \Delta B^2 \geq |\,\langle f|g\rangle\,|^2$$
$$\Rightarrow \Delta A^2 \Delta B^2 \geq (Im\,\langle f|g\rangle)^2$$
$$\Rightarrow \Delta A^2 \Delta B^2 \geq (Im(\langle\psi|\,(\hat{A} - \mathbb{E}[A]I)(\hat{B} - \mathbb{E}[B]I)\,|\psi\rangle))^2$$
$$\Rightarrow \Delta A^2 \Delta B^2 \geq (Im(\mathbb{E}[AB] - 2\mathbb{E}[A]\mathbb{E}[B] - +\mathbb{E}[A]\mathbb{E}[B]))^2$$
$$\Rightarrow \Delta A^2 \Delta B^2 \geq (Im(\mathbb{E}[AB] - \mathbb{E}[A]\mathbb{E}[B]))^2$$

Note that $Imz = \frac{1}{2i}(z - z^*)$

$$\Rightarrow \Delta A^2 \Delta B^2 \geq \left(\frac{1}{2i}(\mathbb{E}[AB] - \mathbb{E}[BA])\right)^2$$
$$\Rightarrow \Delta A^2 \Delta B^2 \geq \left(\langle\psi|\,\frac{1}{2i}[\hat{A}, \hat{B}]\,|\psi\rangle\right)^2$$

33 Linear Algebra III

Hold on to your seats, this is advanced linear algebra!

33.1 Tensor

A scalar is a rank-0 tensor (0D).

A vector is a rank-1 tensor (1D).

A matrix is a rank-2 tensor (2D).

A rank-n tensor is an n-dimensional array with n indices.

A tensor is a subset of a vector.

33.2 Tensor Product

Let

$$|v, w\rangle \triangleq |v\rangle \otimes |w\rangle$$

Given vector spaces H_A and H_B, the tensor product $H_A \otimes H_B$ is a new vector space

Properties

1. $|v\rangle \in H_A,\ |w\rangle \in H_B \implies |v\rangle \otimes |w\rangle \in H_{AB}$

2. Distributive law:

$$|v\rangle \otimes (c_0|w_0\rangle + c_1|w_1\rangle) = c_0(|v\rangle \otimes |w_0\rangle) + c_1(|v\rangle \otimes |w_1\rangle)$$
$$(c_0|v_0\rangle + c_1|v_1\rangle) \otimes |w\rangle = c_0(|v_0\rangle \otimes |w\rangle) + c_1(|v_1\rangle \otimes |w\rangle)$$

3. Inner product

$$\begin{aligned}\langle v_1, w_1 \mid v_2, w_2\rangle &= ((\langle v_1| \otimes \langle w_1|)(|v_2\rangle \otimes |w_2\rangle) \\ &= \langle v_1|v_2\rangle\langle w_1|w_2\rangle\end{aligned}$$

4. $\forall\, |\psi\rangle \in \mathcal{H}_{AB},\ |\psi\rangle = \sum_\mu c_\mu |v_\mu\rangle \otimes |w_\mu\rangle$
   for some sequence of $(c_\mu, |v_\mu\rangle, |w_\mu\rangle)$

Note

1. $|v\rangle \otimes |w\rangle \neq |w\rangle \otimes |v\rangle$
2. $a\,|w\rangle \otimes |v\rangle = |w\rangle \otimes a\,|v\rangle = a(|w\rangle \otimes |v\rangle)$
3. $0 \otimes |v\rangle = |w\rangle \otimes 0 = 0$
4. $\langle v_i, w_j|v_k, w_l\rangle = \langle v_i|v_k\rangle\,\langle w_j|w_l\rangle = \delta_{ik}\delta_{jl}$

The dimensions are multiplied

$$dim(H_A \otimes H_B) = dim(H_A) \times dim(H_B)$$

$$\dim(\underbrace{\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \cdots \otimes \mathbb{C}^2}_{n}) = 2^n \quad \text{(dim of } n\text{-qubit Hilbert space)}$$

$$\{|v_i\rangle \otimes |w_j\rangle\,; 1 \leq i \leq dim(H_A), 1 \leq j \leq dim(H_B)\}$$

is an orthonormal basis for $H_A \otimes H_B$.

33.3 Kronecker Product

The Kronecker product is a way to compute the tensor product of two matrices. It outputs a matrix as well.

Let there be a spin-1/2 system and two particles living H_A and H_B respectively which spans $|0\rangle_A, |1\rangle_A$ and $|0\rangle_B, |1\rangle_B$ respectively. The tensor product basis for $H_A \otimes H_B$ is

$$\{|0\rangle_A \otimes |0\rangle_B, |0\rangle_A \otimes |1\rangle_B, |1\rangle_A \otimes |0\rangle_B, |1\rangle_A \otimes |1\rangle_B\}$$

This can be rewritten as

$$|00\rangle \triangleq |0\rangle_A \otimes |0\rangle_B$$
$$|01\rangle \triangleq |0\rangle_A \otimes |1\rangle_B$$
$$|10\rangle \triangleq |1\rangle_A \otimes |0\rangle_B$$
$$|11\rangle \triangleq |1\rangle_A \otimes |1\rangle_B$$

Let $|0\rangle, |1\rangle$ be a rank-1 tensor, i.e., a vector. Recall from ket states and quantum systems that it is defined as

$$|0\rangle = \begin{pmatrix}1\\0\end{pmatrix} \quad |1\rangle = \begin{pmatrix}0\\1\end{pmatrix}$$

Note that

$$|0\rangle_A = \begin{pmatrix}1\\0\end{pmatrix} \quad |0\rangle_B = \begin{pmatrix}1\\0\end{pmatrix}$$

But when we compute the tensor product (for matrices this is known as the Kronecker product), we get

$$|0\rangle_A \otimes |0\rangle_B = \begin{pmatrix}1\\0\end{pmatrix} \otimes \begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}1\cdot\begin{pmatrix}1\\0\end{pmatrix}\\0\cdot\begin{pmatrix}1\\0\end{pmatrix}\end{pmatrix} = \begin{pmatrix}1\\0\\0\\0\end{pmatrix}$$

We can see the full basis of the new Hilbert space $H_A \otimes H_B$ as

$$|00\rangle = \begin{pmatrix}1\\0\\0\\0\end{pmatrix}, \quad |01\rangle = \begin{pmatrix}0\\1\\0\\0\end{pmatrix}, \quad |10\rangle = \begin{pmatrix}0\\0\\1\\0\end{pmatrix}, \quad |11\rangle = \begin{pmatrix}0\\0\\0\\1\end{pmatrix}$$

33.4 Tensor Product Space

Think of it like a vector space for tensors.

Just like how a vector space is a set of vectors that is closed under addition and scalar multiplication, a tensor product space is a set of tensors that is closed under addition and scalar multiplication.

It is computed by using the tensor product.

33.5 Baker Campbell Hausdorff Formula

Intelligent set of individuals who worked out that

$$e^A e^B = exp(A + B + \frac{1}{2}[A, B] + \frac{1}{12}[A, [A, B]] - \frac{1}{12}[B, [A, B]] + \cdots)$$

if $[A, B] = 0$ does not commute, then $e^A e^B = e^{A+B}$

33.6 Cross Product

Works on 3-element rank-1 tensors i.e., 3D vectors.

$$\vec{A} \times \vec{B} = \begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\A_x & A_y & A_z\\B_x & B_y & B_z\end{vmatrix} = (A_y B_z - A_z B_y)\hat{i} + (A_z B_x - A_x B_z)\hat{j} + (A_x B_y - A_y B_x)\hat{k}$$

34 No-Cloning Theorem

There exists no unitary operator U that can clone a qubit. Note that $\otimes$ is the tensor product.

$$C(|\psi\rangle) = |\psi\rangle \otimes |\psi\rangle$$

Proof

Let

$$C|0\rangle \to |0\rangle|0\rangle$$
$$C|1\rangle \to |1\rangle|1\rangle$$

Then

$$\begin{aligned} C|+\rangle &\to= C\frac{|0\rangle + |1\rangle}{2} \\ &\neq \frac{1}{\sqrt{2}}(|0\rangle|0\rangle + |1\rangle|1\rangle) \\ &\neq |+\rangle|+\rangle \end{aligned}$$

This breaks the linearity of operators.

35 Quantum Information Capacity

A classical bit has 2 states: 0 and 1.

$$\{0, 1\}$$

A qubit has infinite distinct states. Linear combination (superposition) of

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

Example

We can let there be three states in spin-1/2.

$$|z+\rangle \quad \text{british coming by land}$$

$$|z-\rangle \quad \text{british coming by sea}$$

$$|x+\rangle \quad \text{british not coming}$$

So i can measure S_z to be either $+\hbar/2$ or $-\hbar/2$. If it is $+\hbar/2$, then the british are not coming by sea. If it is $-\hbar/2$, then the british are not coming by land.

I can also measure S_x to be either $+\hbar/2$ or $-\hbar/2$. If it is $+\hbar/2$, we don't get any extra information as it could equally be either three. If it is $-\hbar/2$, then the british are coming either by land or sea.

How reliably can we distinguish between a set of quantum states that serve as signals?

Consider Alice and Bob. Alice sends quantum state to Bob and this state is living in some Hilbert space H

$$|\psi\rangle \in H \quad dim(H) = d$$

Alice encodes N equally likely messages

$$\{|\alpha\rangle\}_{\alpha=1}^{N}$$

Bob makes measurements in some basis

$$\{|k\rangle\}_{k=1}^{d'}$$

Bob measures a larger Hilbert space H' because he doesn't know the Hilbert space Alice is using.

$$H \subset H' \quad dim(H') = d' > d$$

Bob still needs to make a determination. But he still needs to make a decision on which of the N messages sent.

Let

$$k = 1, 2, 3, 4, \cdots, d' \quad d' > N$$

for some of these outcomes you can get maybe you assign some like $1, 2$ to be message 1 C_1 and 3 to message 2 C_2, etc. then you need to cover all possibilities of the message Alice sends. You want to map the outcome to messages. Some of the bins may be empty.

The success probability for Bob to correctly distinguish the different messages.

$$P_s = \sum_{\alpha=1}^{N} P(\alpha) \sum_{k \in C_\alpha} P(outcome\ k|\text{message } \alpha \text{ was sent})$$

Recall Born's rule that

$$= \sum_{\alpha=1}^{N} \frac{1}{N} \sum_{k \in C_\alpha} |\langle k|\alpha\rangle|^2$$

We want to try to upperbound this. Let $\{|\phi_n\rangle\}_{r=1}^{d}$ be some orthonormal basis of H

We know that

$$\Pi |\alpha\rangle = |\alpha\rangle$$

And also

$$(\Pi |\alpha\rangle)^\dagger = |\alpha\rangle^\dagger$$
$$\langle\alpha| \Pi^\dagger = \langle\alpha|$$
$$\langle\alpha| \Pi = \langle\alpha|$$

So

$$P_s = \frac{1}{N}\sum_{\alpha}\sum_{k\in C_\alpha} \langle\alpha|k\rangle \langle k|\alpha\rangle$$
$$= \frac{1}{N}\sum_{\alpha}\sum_{k\in C_\alpha} \langle\alpha|\Pi|k\rangle \langle k|\Pi|\alpha\rangle$$
$$= \frac{1}{N}\sum_{\alpha}\sum_{k\in C_\alpha} \langle\alpha| (\Pi |k\rangle \langle k| \Pi) |\alpha\rangle$$
$$= \frac{1}{N}\sum_{\alpha}\sum_{k\in C_\alpha} tr(\langle\alpha| \Pi |k\rangle \langle k| \Pi |\alpha\rangle)$$

Using trace cyclic property

$$= \frac{1}{N}\sum_{\alpha}\sum_{k\in C_\alpha} tr(|\alpha\rangle \langle\alpha| \cdot \Pi |k\rangle \langle k| \Pi)$$
$$\leq \frac{1}{N}\sum_{\alpha}\sum_{k\in C_\alpha} tr(\Pi |k\rangle \langle k| \Pi)$$
$$\leq \frac{1}{N}\sum_{\alpha}\sum_{k\in C_\alpha} tr(\Pi^2 \cdot |k\rangle \langle k|)$$
$$\leq \frac{1}{N} tr(\Pi^2 \sum_{\alpha}\sum_{k\in C_\alpha} |k\rangle \langle k|)$$
$$\leq \frac{1}{N} tr(\Pi^2 I)$$
$$\leq \frac{1}{N} tr(\Pi)$$

Using trace cyclic property and the fact that $tr(\langle\phi_r|\phi_r\rangle) = \langle\phi_r|\phi_r\rangle$ as it is a 1×1 matrix.

$$\leq \frac{1}{N} tr(\sum_{r=1}^{d} |\phi_r\rangle \langle\phi_r|)$$
$$\leq \frac{1}{N} \sum_{r=1}^{d} tr(\langle\phi_r|\phi_r\rangle)$$
$$\leq \frac{1}{N} \sum_{r=1}^{d} \langle\phi_r|\phi_r\rangle$$
$$\leq \frac{1}{N} \sum_{r=1}^{d} 1$$
$$\leq \frac{d}{N}$$

This means the probability of a successful decoding is at most the number of dimensions of the Hilbert space divided by the number of possible messages Alice can send

36 Basic Decoding Theory

This is based on quantum information capacity. Given that Alice is trying to send a message to Bob, the decoding error is

$$P_E = 1 - P_s \geq 1 - \frac{d}{N}$$

Note that

1. If $N \geq d$, then $P_E \geq 0$ – there is no way to reliably distinguish messages
2. lower bound in theorem also holds even if alice sends mixed states $\{\rho_\alpha\}_{\alpha=1}^N$
3. A quantum system with Hilbert space dimension d has information capacity of $\log_2(d)$ bits. This is the maximum number of bits that can be stored, communicated, and read reliably. This is the same amount of infromation from classical d-state system.

37 Basic Distinguishability Theory

This refines basic decoding theory.

Probability of successfully distinguishing between two equally likely pure states $|\alpha_0\rangle, |\alpha_1\rangle$ where $\langle\alpha_0|\alpha_1\rangle = \cos\theta$ is

$$P_s \leq \frac{1}{2}(1 + \sin\theta)$$

Example I

We want to find the optimal measurement to distinguish between two equally likely pure states $|\alpha_0\rangle, |\alpha_1\rangle$ where $\langle\alpha_0|\alpha_1\rangle = \cos\theta$. We can try to do

$$|\alpha_0\rangle = |0\rangle \quad |\alpha_1\rangle = |+\rangle$$
$$\cos\theta = \sin\theta = \frac{1}{\sqrt{2}}$$

and there are is a $|1\rangle$ basis that is ortho to $|0\rangle$. If we measure and get $|1\rangle$ it couldn't have come from $|\alpha_0\rangle$

But it isn't optimal

Example II

The optimal way is to get the bisection angle between $|\alpha_1\rangle, |\alpha_0\rangle$

$$P_s(\tau) = P(\alpha_0)P(\text{measure } k_0 \text{ given } |\alpha_0\rangle) + P(\alpha_1)P(\text{measure } k_1 \text{ given } |\alpha_1\rangle)$$
$$= \frac{1}{2}\cos^2(\tau) + \frac{1}{2}\cos^2(\frac{\pi}{2} - \tau - \theta)$$

Optimize to get

$$\tau^* = \frac{1}{2}\left(\frac{\pi}{2} - \theta\right)$$
$$\Rightarrow \max P_s(\tau) = \frac{1}{2}\cos^2\left(\frac{\pi}{4} - \frac{\theta}{2}\right) + \frac{1}{2}\cos^2(\frac{\pi}{4} - \frac{\theta}{2})$$
$$= \frac{1}{2}(1 + \sin\theta)$$

38 Quantum Cryptography

38.1 Classical Cryptography

How do you communicate securely over a public channel?

Using a shared secret key! (Symmetric key encryption)

Let ciphertext C, plaintext P, and key K be strings of bits $\{0,1\}^*$. Let E be an encryption function and D be a decryption function.

$$C = E(P, K)$$
$$P = D(C, K)$$

There are three people Alice, Bob and Eve where Alice is trying to send data to Bob over a public channel and Eve is listening to it and maliciously trying to obtain the data.

How do we securely share the secret key K in the first place? Classically we use RSA or Diffie-Hellman, but this relies on assumptions about hardness of a computational problem, which may not be hard for a quantum coputer.

In the quantum world, we use quantum key distribution (QKD) to securely share the secret key K without Eve being able to intercept it.

38.2 BB84 Protocol

This is a type of quantum key distribution protocol. It is named after Bennett and Brassard in 1984.

Alice chooses a basis b (subset of the Pauli matrices) and a bit p (i.e., parent)

$$b \in \{X, Z\} \quad p \in \{0, 1\}$$

We know there are an infinite number of possible bases in a qubit. There exists three mutually unbiased bases in a qubit $\{Z, X, Y\}$. BB84 chooses $\{X, Z\}$ as part of the definition of the protocol.

Because a qubit is $d = 2$ by definition, we can only choose from 2 parents.

Alice then creates a qubit state $|\psi_{b,p}\rangle$ to send to Bob. BB84 defines the following mapping:

basis b	parent p	qubit state $\lvert\psi_{b,p}\rangle$
Z	0	$\lvert 0\rangle$
Z	1	$\lvert 1\rangle$
X	0	$\lvert +\rangle = \frac{1}{\sqrt{2}}(\lvert 0\rangle + \lvert 1\rangle)$
X	1	$\lvert -\rangle = \frac{1}{\sqrt{2}}(\lvert 0\rangle - \lvert 1\rangle)$

Example

To generate a shared key, Alice chooses a string of random bases $\in \{X, Z\}^n$ and parents $\in \{0, 1\}^n$ to create a string of qubit $|\psi_{b,p}\rangle$ to send to Bob. Where n is the number of bits to send.

basis b	X	Z	Z	X	Z	X	X	Z
parent p	1	0	1	1	0	1	0	1
qubit $\lvert\psi_{b,p}\rangle$	$\lvert -\rangle$	$\lvert 0\rangle$	$\lvert 1\rangle$	$\lvert -\rangle$	$\lvert 0\rangle$	$\lvert -\rangle$	$\lvert +\rangle$	$\lvert 1\rangle$

Bob chooses a string of random bases $\in \{X, Z\}^n$ and measures the i-th qubit Alice sent in the i-th base.

Bob's basis b'	X	X	Z	Z	X	Z	X	Z
example result	$\|-\rangle$	$\|-\rangle$	$\|1\rangle$	$\|0\rangle$	$\|+\rangle$	$\|1\rangle$	$\|+\rangle$	$\|1\rangle$
or	$\|-\rangle$	$\|+\rangle$	$\|1\rangle$	$\|1\rangle$	$\|-\rangle$	$\|0\rangle$	$\|+\rangle$	$\|1\rangle$
inferred parent (?)	1	1	1	0	0	1	0	1
correct basis	✓		✓				✓	✓
secret key	1		1				0	1

Alice and Bob both announce their basis but not their measurement results. Correct basis for half the qubits. Then use the subsequence as the secret key K.

$$K = 1101$$

Alice and Bob then agree on a random subset $T \subset K$. They publicly announce their bit values only at positions in T.

$$QBER = \frac{\text{number of pos. in } T \text{ where bits differ}}{|T|}$$

If quantum bit error rate QBER is lower than the threshold, then proceed with the remaining bits not in T. In BB84, the threshold is typically around 11% because of Shannon-theoretic entropy.

This is secure against Eve as if Eve intercepts a qubit, Eve can only distinguish among $\{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}$ where error probability $P_E \geq 1 - \frac{d}{N}$ given in basic decoding theory.

If Eve sends qubit along to Bob, they would be wrong half of the time. Alice and Bob can find out in the testing stage. During testing stage, some bits of the key are sent between Alice and Bob and discarded if they do not match. This reduces information that Eve has about the secret key. The probability of the bit disagreeing given Eve intercepted is $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ whereas if Eve doesn't intercept, the probability of the bit disagreeing is 0.

Due to the no-cloning theorem, Eve cannot copy the qubit and send it to Bob.

39 Quantum Dynamics

Quantum dynamics is the study of how quantum systems evolve over time.

39.1 Isolated System

This builds on Schrödinger's equation. State changes from $t_1 \to t_2$ according to some linear operator $U(t_2, t_1)$.

$$|\psi(t_2)\rangle = U(t_2, t_1) |\psi(t_1)\rangle$$

Properties

1. $t_2 = t_1$ gives no change
$$|\psi(t_1)\rangle = U(t_1, t_1) |\psi(t_1)\rangle$$
$$U(t_1, t_1) = I$$

2. composition
$$\begin{aligned} |\psi(t_3)\rangle &= U(t_3, t_2) |\psi(t_2)\rangle \\ &= U(t_3, t_2) U(t_2, t_1) |\psi(t_1)\rangle \\ &= U(t_3, t_1) |\psi(t_1)\rangle \end{aligned}$$
hence
$$U(t_3, t_1) = U(t_3, t_2) U(t_2, t_1)$$

3. inverse
$$I = U(t_1, t_2)U(t_2, t_1)$$
$$U(t_1, t_2) = U^{-1}(t_2, t_1) = U^{\dagger}(t_2, t_1)$$
4. diagonalizable
$$U^{\dagger}U = UU^{\dagger}$$
this means spectral theorem applies. For every orthonormal basis $\{|k\rangle\}$,
$$U = \sum_k \lambda_k |k\rangle \langle k|$$
$$I = \langle k| U^{\dagger}U |k\rangle$$
$$\lambda_k = e^{i\theta_k}$$

39.2 Unitary Evolution

Postulates of an isolated system:

1. $U(t_2, t_1)$ should be independent of initial state.
$$\alpha |\psi\rangle + \beta |\phi\rangle \to U(t_2, t_1)(\alpha |\psi\rangle + \beta |\phi\rangle) =$$
2. must preserve probablity (must sum to 1) if isolated which means U must be unitary if Hilbert space is finite dimensional.
3. assume U is unitary in infinite dimensional Hilbert space.

Unpacking 2. we see that
$$\langle\psi|\psi\rangle = 1$$
This is because Born's rule says that, over $|k\rangle$ basis states,
$$\sum_k |\langle k|\psi\rangle|^2 = 1$$
as probability must sum to 1. Using this lemma, we get
$$\sum_k |\langle k|\psi\rangle|^2 = \sum_k \langle\psi|k\rangle \langle k|\psi\rangle = \langle\psi| \left(\sum_k |k\rangle \langle k|\right) |\psi\rangle$$
$$= \langle\psi| I |\psi\rangle = \langle\psi|\psi\rangle$$
so if we evolve it over time then
$$(U|\psi\rangle)^{\dagger}U|\psi\rangle = 1 \quad \forall \quad |\psi\rangle$$
$$\Rightarrow \quad \langle\psi| U^{+}U |\psi\rangle = 1$$
$$\Rightarrow \quad U^{+}U = I$$

This means that U is unitary.

Note that
$$U(t_0, t) = U^{-1}(t, t_0) = U^{\dagger}(t, t_0)$$

39.3 Schrödinger Equation, Hamiltonian II

Let's derive the Schrödinger equation from the postulates of an isolated system.

Let t_0 be the initial time and t be the time after some time has passed.

Let

$$|\psi(t)\rangle = U(t,t_0)|\psi(t_0)\rangle$$

So

$$\frac{\partial}{\partial t}|\psi(t)\rangle = \frac{\partial U(t,t_0)}{\partial t}|\psi(t_0)\rangle$$

Due to its inverse property where $U(t_0,t)U(t,t_0) = I$,

$$= \frac{\partial U(t,t_0)}{\partial t}\, U(t_0,t)|\psi(t)\rangle$$

$$= \underbrace{\frac{\partial U(t,t_0)}{\partial t}\, U^\dagger(t,t_0)}_{\Lambda(t,t_0)}\, |\psi(t)\rangle$$

Properties

1. $\Lambda(t,t_0)$ is anti-Hermitian.
   Note:

$$\Lambda^\dagger(t,t_0) = U(t,t_0)\,\frac{\partial U^\dagger(t,t_0)}{\partial t}$$

   Differentiate $U(t,t_0)U^\dagger(t,t_0) = I$:

$$\frac{\partial}{\partial t}\big[U(t,t_0)U^\dagger(t,t_0)\big] = \frac{\partial U(t,t_0)}{\partial t}U^\dagger(t,t_0) + U(t,t_0)\frac{\partial U^\dagger(t,t_0)}{\partial t} = 0$$

$$\Rightarrow \quad \Lambda(t,t_0) + \Lambda^\dagger(t,t_0) = 0$$

$$\Rightarrow \quad \Lambda = -\Lambda^\dagger \;\text{ is anti-Hermitian.}$$

2. $\Lambda(t,t_0)$ is independent of t_0

$$\Lambda(t,t_0) = \frac{\partial U(t,t_0)}{\partial t}U^\dagger(t,t_0)$$

$$= \frac{\partial U(t,t_0)}{\partial t}U(t_0,t_1)U^\dagger(t_0,t_1)U^\dagger(t,t_0)$$

$$= \frac{\partial}{\partial t}[U(t,t_0)U(t_0,t_1)][U(t,t_0)U(t_0,t_1)]^\dagger$$

$$= \frac{\partial U(t,t_1)}{\partial t}U^\dagger(t,t_1) = \Lambda(t,t_1)$$

   For any t_0, t_1

Let

$$\Lambda(t) \triangleq \Lambda(t,t_0)$$

$$\Rightarrow \frac{d}{dt}\,|\psi(t)\rangle = \frac{\partial}{\partial t}U(t,t_0)U^\dagger(t,t_0)|\psi(t)\rangle$$

$$= \Lambda(t) |\psi(t)\rangle$$

Then, let

$$\boxed{H(t) \triangleq i\hbar\Lambda(t) = i\hbar\frac{\partial U(t,t_0)}{\partial t}U^\dagger(t,t_0)}$$

This is known as the Hamiltonian, which has the units of energy. In natural units where $\hbar = 1$, it is simply inverse time s^{-1}.

We arrive at the Schrödinger equation:

$$\boxed{i\hbar\frac{d}{dt}|\psi(t)\rangle = H(t)|\psi(t)\rangle}$$

Remarks

1. Derived Schrödinger equation from only the assumption of unitary time evolution. The Hamiltonian operator is derived from $U(t, t_0)$
2. Converse is true where we can construct $U(t, t_0)$ from knowledge of $H(t)$
3. in many physical systems, it easier to find $H(t)$ than finding $U(t, t_0)$

39.4 Example of Finding Hamiltonian

Let there be a particle in space where energy is kinetic energy plus potential energy.

In classical physics, we know that

$$KE = \frac{1}{2}mv^2 = \frac{p^2}{2m}$$

where KE is kinetic energy, m is mass, v is velocity and $p = mv$ is momentum. We know that potential energy $V(x)$ is a function of position x.

Hence in classical physics, energy is

$$E = KE + V(x) = \frac{p^2}{2m} + V(x)$$

A wavelength λ travelling in the direction x can be written as e^{ikx} where wave number k is defined as

$$k \triangleq \frac{2\pi}{\lambda}$$

de Broglie in 1924 hypothesized that every particle has wave-like behavior with wavelength λ

$$\lambda = \frac{h}{p}$$

$$\Rightarrow p = \frac{h}{2\pi/k} = \hbar k$$

But we want this as an operator where the plane wave is an <u>eigenstate</u> of $\hat{p}$ with eigenvalue $\hbar k$ (its momentum).

$$\hat{p}e^{ikx} = \hbar k e^{ikx}$$

Now differentiate e^{ikx} w.r.t x

$$\Rightarrow \frac{\partial}{\partial x}e^{ikx} = ike^{ikx}$$

Multiply by $\hbar/i$

$$\Rightarrow \frac{\hbar}{i}\frac{\partial}{\partial x}e^{ikx} = \hbar k e^{ikx}$$

Comparing like terms we find

$$\hat{p} = \frac{\hbar}{i}\frac{\partial}{\partial x}$$

Anyways, after doing all of that, we're going to guess the Hamiltonian. Guess that the Hamiltonian is $H(t) = E$

$$H(t) = \frac{\hat{p}^2}{2m} + V(\hat{x})$$

Expand the state into basis via spectral decomposition

$$|\psi(t)\rangle = \sum_k \langle x_k|\psi(t)\rangle |x_k\rangle = \int dx \, \langle x|\psi(t)\rangle |x\rangle$$

The intergral is the continuum limit of the discretized version on the left. Let's take the Schrödinger equation and apply it to the state.

$$\langle x| i\hbar\frac{d}{dt}|\psi(t)\rangle = \langle x| H(t) |\psi(t)\rangle$$

LHS and RHS are both manipulated to get

$$i\hbar\frac{\partial}{\partial t}\langle x|\psi(t)\rangle = H(t)\langle x|\psi(t)\rangle$$

$$\Rightarrow i\hbar\frac{\partial}{\partial t}\langle x|\psi(t)\rangle = \langle x| \frac{\hat{p}^2}{2m}|\psi(t)\rangle + V(x)\langle x|\psi(t)\rangle$$

$$\Rightarrow i\hbar\frac{\partial}{\partial t}\langle x|\psi(t)\rangle = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\langle x|\psi(t)\rangle + V(x)\langle x|\psi(t)\rangle$$

$$\Rightarrow i\hbar\frac{\partial}{\partial t}\langle x|\psi(t)\rangle = \left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)\right)\langle x|\psi(t)\rangle$$

If we solve this for specific systems and check against measurements, it will confirm whether or not our guess for $H(t)$ was right

39.5 Wavefunction

Where $|\psi(t)\rangle$ is a state in the Hilbert space, $\psi(x,t)$ is the wavefunction w.r.t time t and position x.

Let

$$\psi(x,t) \triangleq \langle x|\psi(t)\rangle$$

39.6 Uniform dynamics

This is a time-indepndent Hamiltonian $H(t) = H$.

We set an ODE for U which we derived earlier.

$$i\hbar\frac{dU}{dt} = HU \iff \frac{dU}{dt} = -\frac{iH}{\hbar}U$$

If we try it for a scalar value, it is similar to

$$\frac{dy}{dt} = ky(t) \implies y(t) = e^{kt}y(0)$$

Recall matrix exponential:

Guess that $U(t, t_0) = e^{-iHt/\hbar}U_0$

Check:

$$\frac{dU}{dt} = -\frac{iH}{\hbar}e^{-iHt/\hbar}U_0 = -\frac{iH}{\hbar}U$$

Hence true.

Note U_0 is some integration constant which we haven't defined yet. We find U_0

$$I = U(t_0, t_0) = e^{-iHt_0/\hbar}U_0 \implies U_0 = e^{iHt_0/\hbar}$$

$$\implies U(t, t_0) = e^{-iHt/\hbar}e^{iHt_0/\hbar}$$

via Baker Campbell Hausdorff Formula, since $[H, H] = 0$ (trivially as any operator commutes with itself)

$$U(t, t_0) = e^{-iHt/\hbar}e^{iHt_0/\hbar}$$

$$\Rightarrow \boxed{U(t, t_0) = e^{-iH(t-t_0)/\hbar}}$$

Hence this solves SE

$$i\hbar\frac{d}{dt}U(t, t_0) = HU(t, t_0)$$

39.7 Heisenberg Picture

Let observable $\hat{A}$. The Heisen picture operator is

$$\tilde{A}(t) = U^\dagger(t)AU(t)$$

	Schrödinger Picture	Heisenberg Picture
State	$\lvert\psi(t)\rangle = U(t)\lvert\psi(0)\rangle$	$\lvert\psi(0)\rangle$ constant
Observable	A constant	$\tilde{A}(t) = U^\dagger(t)\, A\, U(t)$
Dynamics	$i\hbar\frac{d\lvert\psi\rangle}{dt} = H(t)\lvert\psi(t)\rangle$	$\frac{d\tilde{A}}{dt} = \frac{i}{\hbar}[\tilde{H}(t), \tilde{A}(t)]$

For uniform dynamics, $\tilde{H}(t) = \tilde{H}$. That means $U(t) \triangleq U(t, t_0) = e^{-iHt/\hbar}e^{iHt_0/\hbar}$

We know that something commutes with itself, so $[U(t), H] = 0$

$$\Rightarrow \tilde{H}(t) = U^\dagger(t)HU(t) = H$$

Heisenberg equation of motion for $\tilde{A}(t)$

$$\frac{d}{dt}A(t) = \frac{d}{dt}(U^\dagger AU + U^\dagger A\frac{dU}{dt})$$
$$= -\frac{1}{i\hbar}U^\dagger HAU + \frac{1}{i\hbar}U^\dagger AHU$$
$$= -\frac{i}{\hbar}(U^\dagger HUU^\dagger AU - U^\dagger AUU^\dagger HU)$$

because commutator,

$$\boxed{\frac{d}{dt}\hat{A}(t) = \frac{i}{\hbar}[\hat{H}(t), \hat{A}(t)]}$$

39.8 Ehrenfest Theorem

Given observable $\hat{A}$, recall the expectation value of $\hat{A}$ is

$$\mathbb{E}[A(t)] = \langle\psi(t)|\, A(t)\, |\psi(t)\rangle$$

If we differentiate w.r.t time

$$\frac{d}{dt}\mathbb{E}[A(t)] = \left(\frac{d}{dt}\langle\psi(t)|\right) A(t)\,|\psi(t)\rangle + \langle\psi(t)|\, A(t)\left(\frac{d}{dt}|\psi(t)\rangle\right)$$

$$= -\frac{1}{i\hbar}\langle\psi|\, HA\,|\psi\rangle + \langle\psi|\, A\frac{1}{i\hbar}H\,|\psi\rangle$$

because commutator,

$$= \frac{i}{\hbar}\langle\psi|\,[H,A]\,|\psi\rangle$$

Ehrenfest theorem

$$\boxed{\frac{d}{dt}\mathbb{E}[A(t)] = \frac{i}{\hbar}\mathbb{E}\left[[H,A]\right]}$$

Remarks

1. If $[H,A]=0$, then $\frac{d}{dt}\mathbb{E}[A(t)]=0$ The expected value of A is conserved if A and H are compatible. For example, energy H is always conserved in an isolated system!
2. Sometimes convientn to change basis to Heisenberg picture to study observable dynamics

39.9 Hamiltonian Rotation Quirk

Any 2x2 Hermitian Hamiltonian can be decomposed into an identity plus a rotation generator part. This means that every single-qubit Hamiltonian is a rotation. The Pauli basis spans all 2x2 Hermitian matrices.

$$H = \begin{pmatrix} g_0+g_3 & g_1-ig_2 \\ g_1+ig_2 & g_0-g_3 \end{pmatrix} \qquad g_i \in \mathbb{R}$$

$$= g_0 I + g_1 X + g_2 Y + g_3 Z$$

$$= g_0 I + \vec{g}\cdot\vec{\sigma} \qquad \vec{g} = (g_1, g_2, g_3) = g\hat{n}$$

$$= g_0 I + g\,\hat{n}\cdot\vec{\sigma} \qquad \vec{\sigma} = (X,Y,Z) = (\sigma_x, \sigma_y, \sigma_z)$$

$$= g_0 I + \frac{2g}{\hbar}\,\hat{n}\cdot\vec{S}$$

Remarks

1. If and only B is Hermitian then $U = e^{iB}$ is unitary.
2. Spin/angular momentum generates rotations
3. Unitary V is a dynamical symmetry of system if it commutes with time evolution
$$[V, U(t)] = 0 \Rightarrow [V, H(t)] = 0$$
4. Hermitian A is a conserved quantity if $[A, H(t)] = 0$. This also means $V(\theta) = e^{i\theta A}$ is a dynamical symmetry of the system. A conserved quantity generates dynamical symmetries.

39.10 Time Energy Uncertainty

Relies on quantum dynamics.

Time is not an observable.

We could count N full waves of frequency ω over a time window T. You can only count to within ± 1 wave, so $\Delta N \geq 1$.

The uncertainty in the wavecount is ΔN

We know that

$$N = \frac{\omega T}{2\pi} \quad \Rightarrow \quad \Delta N = \frac{\Delta\omega T}{2\pi} \geq 1 \quad \Rightarrow \quad \Delta\omega T \geq 2\pi$$

From (Planck-Einstein relation),

$$\Delta E \cdot T \geq h = \frac{\hbar}{2}$$

39.11 Example of Quantum Dynamics

An electron is in an magnetic field in the $z+$ direction. There is a particle in state

$$|\psi(0)\rangle = \begin{pmatrix} \cos\theta_0/2 \\ e^{i\phi_0}\sin\theta_0/2 \end{pmatrix}$$

Find out the expected value of S_z, S_x, S_y at time t.

Classical charged particle has

$$\vec{\mu} = \frac{q}{2m}\vec{s}$$

A quantum one also has a g-factor

$$\vec{\mu} = g\frac{q}{2m}\vec{s}$$

In the case of an electron, $q = e, m = m_e, g \approx 2$

$$\vec{\mu} = g\underbrace{\frac{e\hbar}{2m_e}}_{\mu_B}\frac{1}{\hbar}\vec{s}$$

Where μ_B is the Bohr magneton.

The spin in magnetic field $\vec{B}$ experiences a torque.

Clasically, the energy of a magnetic dipole in a B field is

$$U = -\vec{\mu}\cdot\vec{B}$$

In quantum mechanics, the energy is the Hamiltonian operator $\hat{H}$ Let

$$\gamma = \frac{g\mu_B}{\hbar}$$

So

$$\hat{H} = -\vec{\mu}\cdot\vec{B} = -\gamma\vec{s}\cdot\vec{B}$$

Since B field is in the $z+$ direction, $\vec{B} = B\hat{S}_z$

$$\hat{H} = -\gamma B \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = -\gamma B \hat{S}_z$$

To do it the Schrödinger equation way, take the eigenvalues and eigenstates of H

$$H |z\pm\rangle = \mp\gamma B \frac{\hbar}{2} |z\pm\rangle$$

Because Hamiltonian is time-independent, we can use the uniform dynamics solution to find $U(t, t_0)$

$$U(t,0) |z\pm\rangle = e^{-iHt/\hbar} |z\pm\rangle = e^{\pm i\gamma Bt/2} |z\pm\rangle$$

Looking at the state $|\psi(0)\rangle$, we can write it as a superposition of the eigenstates of H

$$|\psi(0)\rangle = \cos\theta_0/2 \, |z+\rangle + e^{i\phi_0} \sin\theta_0/2 \, |z-\rangle$$

$$|\psi(t)\rangle = U(t) \, |\psi(0)\rangle$$

$$|\psi(t)\rangle = (e^{i\gamma Bt/2}) \cos\theta_0/2 \, |z+\rangle + (e^{-i\gamma Bt/2}) e^{i\phi_0} \sin\theta_0/2 \, |z-\rangle$$

Global phase should be separated out

$$|\psi(t)\rangle = e^{i\gamma Bt/2} \left[\cos\theta_0/2 \, |z+\rangle + (e^{i(-\gamma Bt + \phi_0)}) \sin\theta_0/2 \, |z-\rangle \right]$$

Hence

$$\langle\psi(t)|\hat{S}_z|\psi(t)\rangle = \frac{\hbar}{2} \begin{pmatrix} \psi_+^* & \psi_-^* \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix}$$

$$= \frac{\hbar}{2} \begin{pmatrix} \psi_+^* & \psi_-^* \end{pmatrix} \begin{pmatrix} \psi_+ \\ -\psi_- \end{pmatrix}$$

$$= \frac{\hbar}{2} \left(\psi_+^* \psi_+ - \psi_-^* \psi_- \right) = \frac{\hbar}{2} \left(|\psi_+|^2 - |\psi_-|^2 \right)$$

$$= \frac{\hbar}{2} \left(\cos^2 \frac{\theta_0}{2} - \sin^2 \frac{\theta_0}{2} \right)$$

$$= \frac{\hbar}{2} \cos\theta_0 \quad \text{(Note: unchanging in time)}$$

$$\langle\psi(t)|\hat{S}_x|\psi(t)\rangle = \frac{\hbar}{2} \begin{pmatrix} \psi_+^* & \psi_-^* \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix}$$

$$= \frac{\hbar}{2} \begin{pmatrix} \psi_+^* & \psi_-^* \end{pmatrix} \begin{pmatrix} \psi_- \\ \psi_+ \end{pmatrix}$$

$$= \frac{\hbar}{2} \left(\psi_+^* \psi_- + \psi_-^* \psi_+ \right)$$

Plug in $\psi_+ = e^{+i\gamma Bt/2} \cos(\theta_0/2)$ and $\psi_- = e^{-i\gamma Bt/2} e^{i\phi_0} \sin(\theta_0/2)$:

$$\psi_+^* \psi_- = e^{-i\gamma Bt/2} \cos\frac{\theta_0}{2} \cdot e^{-i\gamma Bt/2} e^{i\phi_0} \sin\frac{\theta_0}{2} = e^{i(\phi_0 - \gamma Bt)} \cos\frac{\theta_0}{2} \sin\frac{\theta_0}{2}$$

$$\psi_-^*\psi_+ = e^{-i(\phi_0-\gamma Bt)}\cos\frac{\theta_0}{2}\sin\frac{\theta_0}{2}$$

$$\psi_+^*\psi_- + \psi_-^*\psi_+ = 2\cos\frac{\theta_0}{2}\sin\frac{\theta_0}{2}\cos(\phi_0-\gamma Bt) = \sin\theta_0\cos(\phi_0-\gamma Bt)$$

$$\Rightarrow \langle\psi(t)|\hat{S}_x|\psi(t)\rangle = \frac{\hbar}{2}\sin\theta_0\cos(\phi_0-\gamma Bt)$$

$$\langle\psi(t)|\hat{S}_y|\psi(t)\rangle = \frac{\hbar}{2}\begin{pmatrix}\psi_+^* & \psi_-^*\end{pmatrix}\begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}\begin{pmatrix}\psi_+\\ \psi_-\end{pmatrix}$$

$$= \frac{\hbar}{2}\begin{pmatrix}\psi_+^* & \psi_-^*\end{pmatrix}\begin{pmatrix}-i\psi_-\\ i\psi_+\end{pmatrix}$$

$$= \frac{\hbar}{2}\left(-i\psi_+^*\psi_- + i\psi_-^*\psi_+\right) = \frac{i\hbar}{2}\left(\psi_-^*\psi_+ - \psi_+^*\psi_-\right)$$

Using the same products as above, with $e^{ix} - e^{-ix} = 2i\sin x$:

$$\psi_-^*\psi_+ - \psi_+^*\psi_- = -2i\cos\frac{\theta_0}{2}\sin\frac{\theta_0}{2}\sin(\phi_0-\gamma Bt) = -i\sin\theta_0\sin(\phi_0-\gamma Bt)$$

$$\Rightarrow \langle\psi(t)|\hat{S}_y|\psi(t)\rangle = \frac{i\hbar}{2}\cdot(-i)\sin\theta_0\sin(\phi_0-\gamma Bt) = \frac{\hbar}{2}\sin\theta_0\sin(\phi_0-\gamma Bt)$$

Another way to do it is to use the Heisenberg picture

$$\frac{d}{dt}\hat{A}(t) = \frac{i}{\hbar}[\hat{H},\hat{A}]$$

Recalling the Pauli matrices and their commutators, we can derive it for spin operators S_z, S_x, S_y.

$$[\hat{S}_x,\hat{S}_y] = i\hbar\hat{S}_z, \quad [\hat{S}_y,\hat{S}_z] = i\hbar\hat{S}_x, \quad [\hat{S}_z,\hat{S}_x] = i\hbar\hat{S}_y$$

For our Hamiltonian $\hat{H} = -\gamma B\hat{S}_z$:

$$\frac{d}{dt}\hat{S}_z(t) = \frac{i}{\hbar}[-\gamma B\hat{S}_z(t),\hat{S}_z(t)] = -\frac{i\gamma B}{\hbar}[\hat{S}_z(t),\hat{S}_z(t)] = 0$$

$$\frac{d}{dt}\hat{S}_x(t) = \frac{i}{\hbar}[-\gamma B\hat{S}_z(t),\hat{S}_x(t)]$$

$$= -\frac{i\gamma B}{\hbar}[\hat{S}_z(t),\hat{S}_x(t)]$$

$$= -\frac{i\gamma B}{\hbar}\cdot i\hbar\hat{S}_y(t) = \gamma B\hat{S}_y(t)$$

$$\frac{d}{dt}\hat{S}_y(t) = \frac{i}{\hbar}[-\gamma B\hat{S}_z(t),\hat{S}_y(t)]$$

$$= -\frac{i\gamma B}{\hbar}[\hat{S}_z(t), \hat{S}_y(t)]$$

$$= -\frac{i\gamma B}{\hbar} \cdot (-i\hbar\hat{S}_x(t)) = -\gamma B\hat{S}_x(t)$$

Collecting all three:

$$\frac{d}{dt}(\hat{S}_x, \hat{S}_y, \hat{S}_z) = \gamma B(\hat{S}_y, -\hat{S}_x, 0)$$

Using cross product formula:

$$= -\gamma(0, 0, B) \times (\hat{S}_x, \hat{S}_y, \hat{S}_z)$$

$$\boxed{\frac{d\vec{S}}{dt} = -\gamma\vec{B} \times \vec{S}} \quad \text{(Larmor precession)}$$

40 Composite Systems

40.1 Composite System

A quantum system made up of two or more sub quantum systems. Let H_A and H_B be two state spaces. The state space of the composite system is

$$H_{AB} = H_A \otimes H_B$$

It is represented by a tensor product space.

40.2 States on a Composite System

Recall all the way back to what a quantum state is.

A composite quantum state is a state that lives in a composite system. Recall tensor product of two Hilbert spaces H_A and H_B. A state on a composite system can be written as a sum of tensor products of states on the individual subsystems.

$$|\psi\rangle = \sum_{ij} a_{ij} |v_i\rangle \otimes |w_j\rangle \quad \in H_A \otimes H_B$$

Another way to notationally write this is

$$\boxed{\left|\psi^{(AB)}\right\rangle = \sum_{ij} a_{ij} |v_i\rangle_A \otimes |w_j\rangle_B}$$

The following is a product state. For some $|v^*\rangle \in H_A$ and $|w^*\rangle \in H_B$, a product state can be written as a tensor product of the two states

$$|\psi\rangle = |v^*\rangle \otimes |w^*\rangle \quad |v^*\rangle \in H_A, |w^*\rangle \in H_B$$

An entangled state is any state $|\psi\rangle$ which is not a product state. This means it is not always possible to assign state vectors to individual subsystems.

Einstein famously called this "Spooky action at a distance."

Example 1

Let $\dim H_A = \dim H_B = 2$. Let

$$|\psi_1\rangle = \frac{1}{2}(|0,0\rangle + |0,1\rangle + |1,0\rangle + |1,1\rangle)$$

Note these are being tensor producted with each other but notation wise, we don't write $\otimes$ for simplicity.

$$= \frac{1}{2}\big(|0\rangle|0\rangle + |0\rangle|1\rangle + |1\rangle|0\rangle + |1\rangle|1\rangle\big)$$

$$= \frac{1}{2}\big(|0\rangle(|0\rangle + |1\rangle) + |1\rangle(|0\rangle + |1\rangle)\big)$$

$$= \frac{1}{2}(|0\rangle + |1\rangle)(|0\rangle + |1\rangle)$$

As $|0\rangle + |1\rangle$ forms a state inside H_A and also H_B, this follows the form

$$|\psi\rangle = |v^*\rangle\,|w^*\rangle \quad |v^*\rangle \in H_A, |w^*\rangle \in H_B$$

Example 2

Let

$$|\psi_2\rangle = \frac{1}{2}(|0,0\rangle + |0,1\rangle + |1,0\rangle - |1,1\rangle)$$

$$= \frac{1}{2}\big(|0\rangle|0\rangle + |0\rangle|1\rangle + |1\rangle|0\rangle - |1\rangle|1\rangle\big)$$

$$= \frac{1}{2}\big(|0\rangle(|0\rangle + |1\rangle) + |1\rangle(|0\rangle - |1\rangle)\big)$$

$$= \frac{1}{2}\big(|0\rangle\,|+\rangle + |1\rangle\,|-\rangle\,\big)$$

This does not follow the form

$$|\psi\rangle = |v^*\rangle\,|w^*\rangle \quad |v^*\rangle \in H_A, |w^*\rangle \in H_B$$

This is not written cleanly as a product of two product states in H_A, H_B respectively. Hence it is entangled.

40.2.1 Checking if a state is entangled

To check if a state is entangled, this is an easy way to do it. Let this be the state that we want to check if it is entangled.

$$a_{11}\,|v_1, w_1\rangle + a_{12}\,|v_1, w_2\rangle + a_{21}\,|v_2, w_1\rangle + a_{22}\,|v_2, w_2\rangle$$

Does it follow in this form

$$(a_1\,|v_1\rangle + a_2\,|v_2\rangle) \otimes (b_1\,|w_1\rangle + b_2\,|w_2\rangle)$$

To have a solution

$$a_{11} = a_1 b_1$$

$$a_{21} = a_2 b_1$$

$$a_{12} = a_1 b_2$$

$$a_{22} = a_2 b_2$$

$$a_{11}a_{22} - a_{12}a_{21} = a_1 b_1 a_2 b_2 - a_1 a_2 a_2 b_1 = 0$$

Must all be true.

Hence

$$\text{product state} \Leftrightarrow \det \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = 0$$

40.3 Observables on a Composite System

Don't forget that operators can sometimes be observables.

Let $F \in L(H_A), G \in L(H_B)$ be operators.

Then $F \otimes G \in L(H_A \otimes H_B)$ is defined as

$$(F \otimes G) |v, w\rangle \triangleq F |v\rangle \otimes G |w\rangle$$

Upgrading/promoting subsystem operators gets us

$$F \in L(H_A) \to F \otimes I \in L(H_A \otimes H_B)$$
$$G \in L(H_B) \to G \otimes I \in L(H_A \otimes H_B)$$

Where L is the set of all linear operators on a Hilbert space. This means that any operator acting on a subsystem H_A can be upgraded to an operator that makes it act on a larger dimension $H_A \otimes H_B$.

This is because

$$(F \otimes I)(I \otimes G) |v, w\rangle = (F \otimes I)(|v\rangle \otimes G |w\rangle) = F |v\rangle \otimes G |w\rangle$$

Conversely,

$$(I \otimes G)(F \otimes I)(|v\rangle \otimes |w\rangle) = F |v\rangle \otimes G |w\rangle$$

Notation wise, let

$$F^{(A)} \triangleq F \otimes I$$
$$G^{(B)} \triangleq I \otimes G$$

This is form is overloaded, so don't get confused with notation of states which also can have an superscript (A).

Note

$$\boxed{[F^{(A)}, G^{(B)}] = [F \otimes I, I \otimes G] = 0}$$

For a composite system $H = H_A \otimes H_B$, an operators acting only on one subsystem extends to the full space by tensoring with the identity on the other.

	Subsystem A	Subsystem B
Operator	$Q^{(A)} = Q \otimes I$	$R^{(B)} = I \otimes R$
Eigenvalues	q_a	r_b
Eigenvectors	$\|a\rangle$	$\|b\rangle$
Index range	$a = 1, 2, \ldots, d_A$	$b = 1, 2, \ldots, d_B$

Sum

$$Q^{(A)} + R^{(B)} = Q \otimes I + I \otimes R$$

- Eigenvectors: $|a\rangle \otimes |b\rangle$
- Eigenvalues: $q_a + r_b$

Product

$$Q^{(A)} R^{(B)} = Q \otimes R$$

- Eigenvectors: $|a\rangle \otimes |b\rangle$
- Eigenvalues: $q_a \, r_b$

40.4 Hamiltonian on a Composite System

40.4.1 Non-interacting Hamiltonian

$$H^{(total)} = H^{(A)} + H^{(B)}$$

Where $H^{(A)}, H^{(B)}$ are non interacting.

Two spin-1/2 particles exist where

$$\begin{aligned} H^{(total)} &= \hat{S}_x^{(total)} + \hat{S}_x^{(A)} + \hat{S}_x^{(B)} \\ &= \hat{S}_x \otimes I + I \otimes \hat{S}_x \end{aligned}$$

Let there be a state where

$$|\psi\rangle = a_{11}|+,+\rangle + a_{12}|+,-\rangle + a_{21}|-,+\rangle + a_{22}|-,-\rangle$$

Recall what a $\hat{S}_x$ operator does <u>here</u>.

$$S_x \triangleq \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

basically,

$$\begin{aligned} \hat{S_x}^{(A)} &= \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ &= \frac{\hbar}{2}\begin{pmatrix} 0 \cdot I & 1 \cdot I \\ 1 \cdot I & 0 \cdot I \end{pmatrix} \\ &= \frac{\hbar}{2}\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \end{aligned}$$

So

$$\begin{aligned} \hat{S}_x^{(A)}|\psi\rangle &= \frac{\hbar}{2}(a_{11}|+,+\rangle + a_{12}|+,-\rangle - a_{21}|-,+\rangle - a_{22}|-,-\rangle) \\ \hat{S}_x^{(B)}|\psi\rangle &= \frac{\hbar}{2}(a_{11}|+,+\rangle - a_{12}|+,-\rangle + a_{21}|-,+\rangle - a_{22}|-,-\rangle) \end{aligned}$$

$$\hat{S}_x^{(total)}|\psi\rangle = \hbar(a_{11}|+,+\rangle - a_{12}|-,-\rangle)$$

Eigenstate of $\hat{S}_x^{(\text{tot})}$	Eigenvalue
$\|+,+\rangle$	$S_x^{(\text{tot})} = \hbar$
$\|+,-\rangle,\ \|-,+\rangle$	$S_x^{(\text{tot})} = 0$
$\|-,-\rangle$	$S_x^{(\text{tot})} = -\hbar$

Looks like a spin-1

40.4.2 Time Evolution of Non-interacting Composite Systems

If $H = \hat{S}_x^{(total)}$ then

$$\begin{aligned} e^{-it\hat{S}_x^{(total)}} &= e^{-it\left(\hat{S}_x^{(A)} + \hat{S}_x^{(B)}\right)} \\ &= e^{-it\hat{S}_x^{(A)}} e^{-it\hat{S}_x^{(B)}} \\ &= (e^{-it\hat{S}} \otimes I)(I \otimes e^{-it\hat{S}}) \\ &= e^{-it\hat{S}_x} \otimes e^{-it\hat{S}_x} \end{aligned}$$

Note that

$$H^{(\text{tot})} = H_1^{(A)} + H_2^{(B)} \implies U^{(\text{tot})} = U_1^{(A)} U_2^{(B)} = U_1 \otimes U_2$$

$$U^{(\text{tot})}(|v\rangle \otimes |w\rangle) = (U_1|v\rangle) \otimes (U_2|w\rangle)$$

$$\text{product state} \longrightarrow \text{product state}$$

Unitary evolution under non-interacting Hamiltonian cannot generate an entangled state.

40.4.3 Interacting Hamiltonian

$$H = \underbrace{H^{(A)} \otimes I + I \otimes H^{(B)}}_{\text{local part}} + \underbrace{H_{\text{int}}}_{\text{interaction term}}$$

Where H_{int} is any operator on $H_A \otimes H_B$.

Example

Note σ_x is defined in Pauli matrix.

$$H^{(total)} = \hbar\omega\sigma_x^{(A)}\sigma_x^{(B)} = \hbar\omega(\sigma_x \otimes \sigma_x)$$

Note that $\sigma_x \otimes \sigma_x$ cannot be written as $\sigma_x \otimes I + I \otimes \sigma_x$ hence this belongs in the H_{int} term. We observe that the unitary evolution is

$$U^{(total)} = e^{-itH/\hbar} = e^{-i\omega t(\sigma_x \otimes \sigma_x)}$$

Suppose $|\psi(0)\rangle = |0\rangle \otimes |0\rangle$

$$|\psi(0)\rangle = \frac{|+\rangle + |-\rangle}{\sqrt{2}} \otimes \frac{|+\rangle + |-\rangle}{\sqrt{2}}$$

$$= \frac{1}{2}\left(|+,+\rangle + |+,-\rangle + |-,+\rangle + |-,-\rangle\right)$$

Then via the unitary time evolution we get

$$\begin{aligned}
|\psi(t)\rangle &= e^{-i\omega t\sigma_x\otimes\sigma_x}|\psi(0)\rangle \\
&= \frac{1}{2}\left(e^{-i\omega t\sigma_x\otimes\sigma_x}|+,+\rangle + e^{-i\omega t\sigma_x\otimes\sigma_x}|+,-\rangle + e^{-i\omega t\sigma_x\otimes\sigma_x}|-,+\rangle + e^{-i\omega t\sigma_x\otimes\sigma_x}|-,-\rangle\right) \\
&= \frac{1}{2}\left(e^{-i\omega t}|+,+\rangle + e^{i\omega t}|+,-\rangle + e^{i\omega t}|-,+\rangle + e^{-i\omega t}|-,-\rangle\right)
\end{aligned}$$

Note

$$e^{-i\omega t\sigma_x\otimes\sigma_x}|+,+\rangle = e^{-i\omega t}|+,+\rangle$$

$$e^{-i\omega t\sigma_x\otimes\sigma_x}|+,-\rangle = e^{+i\omega t}|+,-\rangle$$

$$e^{-i\omega t\sigma_x\otimes\sigma_x}|-,+\rangle = e^{+i\omega t}|-,+\rangle$$

$$e^{-i\omega t\sigma_x\otimes\sigma_x}|-,-\rangle = e^{-i\omega t}|-,-\rangle$$

So

$$\det \frac{1}{2}\begin{pmatrix} e^{-i\omega t} & e^{i\omega t} \\ e^{i\omega t} & e^{-i\omega t} \end{pmatrix} = \frac{1}{4}\left(e^{-2i\omega t} - e^{2i\omega t}\right) = -\frac{i}{2}\sin(2\omega t) \neq 0$$

$$\implies \text{Entangled state for } t > 0, \quad t \neq \frac{k\pi}{\omega}$$

$$\text{Maximal when } t = \frac{\pi}{4\omega}$$

40.5 Measuring a Composite System

For a non-composite system, recall that the set of measurements you can make is the eigenvalues of the observable.

Recall measuring a quantum state.

Entanglement doesn't always require a interacting Hamiltonian. They can also be created by jointly measuring a product state.

Suppose we start with $|\psi\rangle = |0\rangle \otimes |0\rangle$ but measured in the x axis

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \longrightarrow \text{product state}$$

Measure $\hat{S}_x^{(\text{tot})}$, get 0 with $\frac{1}{2}$ probability. Mathematically, we use a projector to measure the state. Via the projection rule we get

$$\frac{1}{\sqrt{1/2}}\Pi_{S_x} = 0 \quad |\psi\rangle = \frac{1}{\sqrt{2}}\left(|+,-\rangle + |-,+\rangle\right)$$

$$\frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \longrightarrow \text{entangled state!}$$

This means that joint measurement can product entanglement. Equivalent to interacting via third party.

Consider a composite system $H_{\text{Alice}} \otimes H_{\text{Bob}}$ with subsystem observables having eigenvalues q_a, r_b and eigenvectors $|a\rangle$, $|b\rangle$.

Joint probability for outcome (q_a, r_b):

$$P(a,b) = \left|\left\langle a, b|\Psi^{(AB)}\right\rangle\right|^2$$

Marginal probabilities:

$$P(a) = \sum_b p(a,b), \qquad P(b) = \sum_a p(a,b)$$

No-signalling from A to B. Expand the joint state in the product basis:

$$\left|\Psi^{(AB)}\right\rangle = \sum_{a,b} c_{ab}\,|a\rangle \otimes |b\rangle\,, \qquad p(a,b) = |c_{ab}|^2$$

Group the sum by b to isolate Bob's basis:

$$\left|\Psi^{(AB)}\right\rangle = \sum_b \underbrace{\left(\sum_a c_{ab}\,|a\rangle\right)}_{|\phi_b\rangle \text{ (unnormalized!)}} \otimes |b\rangle = \sum_b |\phi_b\rangle \otimes |b\rangle$$

With this notation, $P(a,b) = |\langle a|\phi_b\rangle|^2$, so Bob's marginal is

$$P(b) = \sum_a P(a,b) = \sum_a \langle \phi_b|a\rangle\,\langle a|\phi_b\rangle = \langle\phi_b|\phi_b\rangle$$

The middle step used $\sum_a |a\rangle\langle a| = I$ — which holds for *any* orthonormal basis $\{|a\rangle\}$ on Alice's side. Therefore $p(b)$ is independent of the choice of Alice's measurement basis. Alice cannot influence Bob's marginal statistics by switching what she measures.

Suppose we apply $U^{(A)}$ to Alice's subsystem

$$\left|\Psi^{(AB)}\right\rangle = U^{(A)} \otimes I \left|\Psi^{(AB)}\right\rangle = \sum_b U\,|\phi_b\rangle \otimes |b\rangle$$

This means that the marginal probability distributiuon of Bob's outcome becomes

$$P'(b) = \langle\phi_b|\,U^{+}U\,|\phi_b\rangle = \langle\phi_b|\phi_b\rangle = P(b)$$

This means Bob's probabilities are unaffected.

40.6 Neutron Interferometry

Let

$$R_{\hat{n}}(\theta) = e^{-i\theta \hat{J}_{\hat{n}}/\hbar}$$

This is how you rotate a quantum state by an angle θ around $\hat{n}$ where $\hat{J}_{\hat{n}}$ is the angular momentum operator along $\hat{n}$.

We experimentally observed that

$$R_x(2\pi) = e^{-i2\pi \hat{S}_x/\hbar} = -I, \qquad R_x(4\pi) = I$$

Given a $\hat{S}_x$ operation, turning it $\theta = 2\pi$ causes there to be a negative sign infront of the I. Rotating it another 2π causes it to return to the same state. This is definitely not how it works in classical physics, where if you rotate something 2π you'd get right back to where it started. This actually means we live in a 4π-world through experimental verification.

This was tested by using a neutron interferometer. Basically, imagine if a neutron at $|\psi\rangle$ hits a silicon crystal beam splitter. The state of the neutron is its position $\{u, l\}$ tensor product its spin $\{+, -\}$. This is just how we define it.

Basis: $\{|n,+\rangle\,,\ |n,-\rangle\,,\ |\ell,+\rangle\,,\ |\ell,-\rangle\}$

$$|\psi\rangle = |\text{position}\rangle \otimes |\text{spin}\rangle$$

Neutron beam splitter (silicon crystal): $|n,s\rangle$ splits into transmitted $|n,s\rangle$ and reflected $|\ell,s\rangle$.

$$(B \otimes I)\,|n,s\rangle = \tfrac{1}{\sqrt{2}}\big(|n\rangle + |\ell\rangle\big) \otimes |s\rangle$$
$$(B \otimes I)\,|\ell,s\rangle = \tfrac{1}{\sqrt{2}}\big(|n\rangle - |\ell\rangle\big) \otimes |s\rangle$$
$$B = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

The lower path passes through a static magnetic field B, which applies a spin rotation $R_x(\theta)$. The two paths recombine at a second beam splitter, with detectors D_0 and D_1 on the two outputs.

The full evolution is

$$|\psi_{\text{out}}\rangle = (B \otimes I)\,U\,(B \otimes I)\ |u,s\rangle$$

where the magnetic field implements the conditional rotation

$$U = |u\rangle\langle u| \otimes I + |\ell\rangle\langle \ell| \otimes R_x(\theta).$$

Case $\theta = 2\pi$:

$$|u,s\rangle \xrightarrow{B\otimes I} \tfrac{1}{\sqrt{2}}\big(|u\rangle + |\ell\rangle\big) \otimes |s\rangle$$
$$\xrightarrow{R_x(2\pi)=-I} \tfrac{1}{\sqrt{2}}\big(|u\rangle - |\ell\rangle\big) \otimes |s\rangle \xrightarrow{B\otimes I} |\ell,s\rangle$$

$\Rightarrow$ neutron at D_1.

Case $\theta = 4\pi$:

$$|u,s\rangle \longrightarrow |u,s\rangle \quad \Rightarrow \text{ neutron at } D_0.$$

This is fully deterministic as in general, $P(D_0) = \cos^2(\theta/4)$ and $P(D_1) = \sin^2(\theta/4)$, so $\theta = 2\pi, 4\pi$ are the special tunings where one probability hits 1.

This is the physical consequence of $R_{\hat{n}}(2\pi) = -I \neq I$.

This experimentally proves that we live in a 4π-world.

40.7 Bell States

For reference, Bell defined these as

$$\left|\Phi^+\right\rangle = \tfrac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right)$$
$$\left|\Phi^-\right\rangle = \tfrac{1}{\sqrt{2}}\left(|00\rangle - |11\rangle\right)$$
$$\left|\Psi^+\right\rangle = \tfrac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right)$$
$$\left|\Psi^-\right\rangle = \tfrac{1}{\sqrt{2}}\left(|01\rangle - |10\rangle\right)$$

40.8 No Communication Theorem

With access to state

$$\left|\Psi^{(AB)}\right\rangle$$

which may or may not be entangled, Alice cannot convey information to Bob by

1. Choosing a basic measurement on A
2. Choosing a unitary evolution operator for A

But Alice's choice of measurement basis cannot affect Bob's measurement statistics, her measurement outcome CAN affect Bob's state.

Let $|\psi_a\rangle$ be the state of Bob's qubit conditioned on Alice's outcome a.

$$|\psi_a\rangle \triangleq \left\langle a^{(A)}|\Psi^{(AB)}\right\rangle$$

Let a be the outcome of Alice's measurement. If measured in the z axis, it'd be $a \in \{0, 1\}$ and if measured in the x axis, it'd be $a \in \{+, -\}$.

We can decompose the earlier state into

$$\left|\Psi^{(AB)}\right\rangle = \sum_a |a\rangle \otimes |\psi_a\rangle$$

What does this mean? This means the basis $|a\rangle$ cross tensor product with the state itself. This promotes Via Born's rule

$$P(a, b) = |\langle b|\psi_a\rangle|^2$$

Via conditional probability

$$P(b|a) = \frac{1}{P(a)}|\langle b|\psi_a\rangle|^2 = \frac{\langle\psi_a|b\rangle\langle b|\psi_a\rangle}{\langle\psi_a|\psi_a\rangle}$$
$$= |\left\langle b|\hat{\psi}_a\right\rangle|^2$$

Recall magnitude

$$\left|\hat{\psi}_a\right\rangle \triangleq \frac{1}{\sqrt{P(a)}}|\psi_a\rangle = \frac{|\psi_a\rangle}{||\,|\psi_a\rangle\,||} = \frac{|\psi_a\rangle}{\sqrt{\langle\psi_a|\psi_a\rangle}}$$

Note that generally, conditional state can be obtained via a partial inner product.

$$\boxed{\left\langle a^{(A)}|\Psi^{(AB)}\right\rangle \triangleq (\langle a| \otimes I)\left|\Psi^{(AB)}\right\rangle}$$

Note this yields a vector in H_B. Where $\left|\Psi^{(AB)}\right\rangle$ is in $H_A \otimes H_B$

Example:

$$\left|\Psi_-^{(AB)}\right\rangle = \frac{1}{\sqrt{2}}\left(\left|0^{(A)}\right\rangle \otimes \left|1^{(B)}\right\rangle - \left|1^{(A)}\right\rangle \otimes \left|0^{(B)}\right\rangle\right)$$

Suppose Alice measures in z basis

$$\left|\Psi_-^{(AB)}\right\rangle = \frac{1}{\sqrt{2}}\left(\left|0^{(A)}\right\rangle \otimes \left|1^{(B)}\right\rangle - \left|1^{(A)}\right\rangle \otimes \left|0^{(B)}\right\rangle\right)$$

Outcome	conditional state (unnormalized)
$a=0$	$\left\|\psi_0^{(B)}\right\rangle = \left\langle 0^{(A)}\|\Phi_-^{(AB)}\right\rangle = \frac{1}{\sqrt{2}}\left\|1^{(B)}\right\rangle$ $\quad p(a=0)=\frac{1}{2}$
$a=1$	$\left\|\psi_1^{(B)}\right\rangle = \left\langle 1^{(A)}\|\Phi_-^{(AB)}\right\rangle = -\frac{1}{\sqrt{2}}\left\|0^{(B)}\right\rangle$ $\quad p(a=1)=\frac{1}{2}$

We can see that Bob's qubit is projected onto z eigenstate opposite to Alice's outcome.

Let's say that Alice measures in x basis.

Outcome	Conditional state (unnormalized)
$a=+$	$\left\|\psi_+^{(B)}\right\rangle = \left\langle +^{(A)}\|\Phi_-^{(AB)}\right\rangle = \frac{\left\langle 0^{(A)}\right\| + \left\langle 1^{(A)}\right\|}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}\left(\left\|0^{(A)}1^{(B)}\right\rangle - \left\|1^{(A)}0^{(B)}\right\rangle\right) = \frac{\left\|1^{(B)}\right\rangle}{2} - \frac{\left\|0^{(B)}\right\rangle}{2} = -\frac{\left\|-^{(B)}\right\rangle}{\sqrt{2}}$ $p(a=+)=\frac{1}{2}$
$a=-$	$\left\|\psi_-^{(B)}\right\rangle = \left\langle -^{(A)}\|\Phi_-^{(AB)}\right\rangle = \frac{\left\|+^{(B)}\right\rangle}{\sqrt{2}}$ $p(a=-)=\frac{1}{2}$

Bob's qubit is projected to opposite basis state as ALice's measurement Somehow, Alice's choice of measurement basis affects Bob's state. If Alice measures in z basis, Bob's conditional state is $|0\rangle$ or $|1\rangle$ with 50/50 probability. If Alice measures in x basis, Bob's conditional state is $|+\rangle$ or $|-\rangle$ with 50/50 probability.

$$\hat{\rho}_B = \tfrac{1}{2}|0\rangle\langle 0| + \tfrac{1}{2}|1\rangle\langle 1| = \tfrac{1}{2}|+\rangle\langle +| + \tfrac{1}{2}|-\rangle\langle -| = \frac{1}{2}I$$

40.9 Entanglement and Quantum Nonlocality

In quantum mechanics -> if z and x observables are incompaible, getting a definite value for z means losing the ability to know x.

Einstein-Podolsky-Rosen (1935) argued that quantum mehcanics is incomplete as physics should obey principles of local realism whereby

1. locality: measurement results cannot dpeend on what happens far away
2. realism: if something is predictable without disturbance, it must have already been there

EPR postulated that let there be a pair of qubits in $\left|\Psi_-^{(AB)}\right\rangle$. Alice takes one and Bob takes one. They then separate light years from each other. Alice now measures her qubit in the z basis. Let's say she gets 0.

$$\left|\Psi_-^{(AB)}\right\rangle = \frac{1}{\sqrt{2}}\left(\left|0^{(A)}\right\rangle \otimes \left|1^{(B)}\right\rangle - \left|1^{(A)}\right\rangle \otimes \left|0^{(B)}\right\rangle\right)$$

Multiplying by $\left\langle 0^{(A)}\right|$ on the left, we get

$$\left\langle 0^{(A)}|\Psi_-^{(AB)}\right\rangle = \left(\langle 0|^{(A)} \otimes I^{(B)}\right)\left|\Psi_-^{(AB)}\right\rangle$$

$$= \left(\langle 0|^{(A)} \otimes I^{(B)}\right) \cdot \tfrac{1}{\sqrt{2}}\left(\left|0^{(A)}\right\rangle \otimes \left|1^{(B)}\right\rangle - \left|1^{(A)}\right\rangle \otimes \left|0^{(B)}\right\rangle\right)$$

$$= \frac{1}{\sqrt{2}}\left[\left(\langle 0|^{(A)} \otimes I^{(B)}\right)\left(\left|0^{(A)}\right\rangle \otimes \left|1^{(B)}\right\rangle\right) - \left(\langle 0|^{(A)} \otimes I^{(B)}\right)\left(\left|1^{(A)}\right\rangle \otimes \left|0^{(B)}\right\rangle\right)\right]$$
$$= \frac{1}{\sqrt{2}}\left|1^{(B)}\right\rangle$$

That means, via. Born's rule, if Alice observed a 0 outcome, Bob's qubit is in the state $|1\rangle$ with 100% probability.

EPR argues that it breaks local realism as local realism wants Bob's qubit to have definite values of x and z observables.

Another example would be let there be two qualities of a particle: hardness $\{hard, soft\}$ and color $\{black, white\}$ if there is a particle with $(\hat{z}+, \hat{x}+)$ which just means it's like $(soft, white)$. This is not an eigenstate because in quantum mechanics, knowing $\hat{z} = +$ means losing the ability to know $\hat{x} = -$ or $\hat{x} = +$. We're just constructing a model that obeys EPR.

41 Cheat Sheet

$$|a\rangle \triangleq \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} \quad a_0, a_1 \in \mathbb{C}$$

$$|a\rangle\langle a| = \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} \begin{pmatrix} a_0^* & a_1^* \end{pmatrix} = \begin{pmatrix} a_0 a_0^* & a_0 a_1^* \\ a_1 a_0^* & a_1 a_1^* \end{pmatrix} = \begin{pmatrix} |a_0|^2 & a_0 a_1^* \\ a_1 a_0^* & |a_1|^2 \end{pmatrix}$$

$$\langle a|a\rangle = \begin{pmatrix} a_0^* & a_1^* \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = a_0^* a_0 + a_1^* a_1 = |a_0|^2 + |a_1|^2$$

2x2 matrix times 2x2 matrix

$$\begin{pmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \end{pmatrix} \begin{pmatrix} B_{1,1} & B_{1,2} \\ B_{2,1} & B_{2,2} \end{pmatrix} =$$

$$\begin{pmatrix} A_{1,1}B_{1,1} + A_{1,2}B_{2,1} & A_{1,1}B_{1,2} + A_{1,2}B_{2,2} \\ A_{2,1}B_{1,1} + A_{2,2}B_{2,1} & A_{2,1}B_{1,2} + A_{2,2}B_{2,2} \end{pmatrix}$$

3x1 matrix times 1x3 matrix

$$\begin{pmatrix} A_{1,1} \\ A_{2,1} \\ A_{3,1} \end{pmatrix} \begin{pmatrix} B_{1,1} & B_{1,2} & B_{1,3} \end{pmatrix} =$$

$$\begin{pmatrix} A_{1,1}B_{1,1} & A_{1,1}B_{1,2} & A_{1,1}B_{1,3} \\ A_{2,1}B_{1,1} & A_{2,1}B_{1,2} & A_{2,1}B_{1,3} \\ A_{3,1}B_{1,1} & A_{3,1}B_{1,2} & A_{3,1}B_{1,3} \end{pmatrix}$$

1x3 matrix times 3x1 matrix

$$\begin{pmatrix} A_{1,1} & A_{1,2} & A_{1,3} \end{pmatrix} \begin{pmatrix} B_{1,1} \\ B_{2,1} \\ B_{3,1} \end{pmatrix} =$$

$$A_{1,1}B_{1,1} + A_{1,2}B_{2,1} + A_{1,3}B_{3,1}$$

Tranpose, adjoint

$$(A^T)_{i,j} = A_{j,i} \quad (A^\dagger)_{i,j} = A^*_{j,i}$$

Hermitian, Unitary Operators

$$A \text{ is Hermitian} \iff A^\dagger = A$$

$$U \text{ is Unitary} \iff U^\dagger U = UU^\dagger = I$$

Matrix Exponential

For a square matrix X,

$$e^X \triangleq \sum_{n=0}^{\infty} \frac{X^n}{n!}$$

$$e^{At} = I + At + \frac{(At)^2}{2!} + \frac{(At)^3}{3!} + \cdots$$

$$\frac{d}{dt} e^{At} = Ae^{At} = e^{At}A$$

$U(t) = e^{tX}$ is the unique solution to $\frac{dU}{dt} = XU(t)$, $U(0) = I$.

If $X = PDP^{-1}$ with $D = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$,

$$e^X = P\,\mathrm{diag}(e^{\lambda_1}, \ldots, e^{\lambda_n})\,P^{-1}$$

If H is Hermitian, then e^{iH} is unitary:

$$(e^{iH})^\dagger e^{iH} = I$$

Taylor Series

$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} \left.\frac{d^n f}{dx^n}\right|_a (x-a)^n$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

Euler's Formula

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \qquad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

EV, Variance, Standard Deviation

$$\mathbb{E}[X] = \sum_x xP(x)$$

$$Var(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

$$Std(X) = \sqrt{Var(X)}$$

Spin-1/2, Pauli Matrices

This spin is for 2-dimensional Hilbert spaces. Electrons, protons, neutrons and quarks have spin-1/2. For spin-1/2 systems,

$$S_x \triangleq \frac{\hbar}{2}\sigma_x \quad S_y \triangleq \frac{\hbar}{2}\sigma_y \quad S_z \triangleq \frac{\hbar}{2}\sigma_z$$

$$S_x \triangleq \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad S_y \triangleq \frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad S_z \triangleq \frac{\hbar}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$|z+\rangle = |{\uparrow}\rangle = |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |z-\rangle = |{\downarrow}\rangle = |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$|x+\rangle = |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$|x-\rangle = |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$|y+\rangle = \frac{1}{\sqrt{2}}(|z+\rangle + i|z-\rangle) = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ i \end{pmatrix}$$

$$|y-\rangle = \frac{1}{\sqrt{2}}(|z+\rangle - i|z-\rangle) = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -i \end{pmatrix}$$

Spin-1

This spin is for 3-dimensional Hilbert spaces. W/Z bosons, photons and gluons have spin-1. For spin-1 systems,

$$S_x \triangleq \frac{\hbar}{2}\Sigma_x \quad S_y \triangleq \frac{\hbar}{2}\Sigma_y \quad S_z \triangleq \frac{\hbar}{2}\Sigma_z$$

$$S_x \triangleq \frac{\hbar}{\sqrt{2}}\begin{pmatrix}0 & 1 & 0\\ 1 & 0 & 1\\ 0 & 1 & 0\end{pmatrix} \quad S_y \triangleq \frac{\hbar}{\sqrt{2}}\begin{pmatrix}0 & -i & 0\\ i & 0 & -i\\ 0 & i & 0\end{pmatrix}$$

$$S_z \triangleq \hbar\begin{pmatrix}1 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & -1\end{pmatrix}$$

$$|1\rangle = \begin{pmatrix}1\\0\\0\end{pmatrix} \quad |0\rangle = \begin{pmatrix}0\\1\\0\end{pmatrix} \quad |-1\rangle = \begin{pmatrix}0\\0\\-1\end{pmatrix}$$

For photons, S_z is often called rectilinear and S_x is often called diagonal and S_y are often called circular.

Schrödinger Equation

$$i\hbar\frac{d}{dt}|\psi(t)\rangle = H(t)|\psi(t)\rangle$$

Uniform Dynamics

For time independent Hamiltonian $H(t) = H$,

$$\Rightarrow U(t, t_0) = e^{-iH(t-t_0)/\hbar}$$

$$|\psi(t)\rangle = U(t, t_0)|\psi(t_0)\rangle$$

Born's Rule

The probability of observing eigenvalue λ_α of an eigenstate $|\alpha\rangle$ is

$$P_\alpha = |\langle\alpha|\psi\rangle|^2$$

Heisenberg Picture

$$\Delta A \Delta B \geq \frac{1}{2}\left|\mathbb{E}\left[[\hat{A}, \hat{B}]\right]\right|$$

$$\Delta A^2 \Delta B^2 \geq \left|\langle\psi|\frac{1}{2j}[\hat{A}, \hat{B}]|\psi\rangle\right|^2 = \left(\frac{1}{2j}\mathbb{E}\left[[\hat{A}, \hat{B}]\right]\right)^2$$

Ehrenfest Theorem

$$\frac{d}{dt}\mathbb{E}[A(t)] = \frac{i}{\hbar}\mathbb{E}\left[[H, A]\right]$$

Time Energy Uncertainty

$$\Delta E \cdot T \geq h = \frac{\hbar}{2}$$